

- 1 The speed of a 100 metre runner in ms^{-1} is measured electronically every 4 seconds.

The measurements are plotted as points on the speed-time graph in Fig. 6. The vertical dotted line is drawn through the runner's finishing time.

Fig. 6 also illustrates Model P in which the points are joined by straight lines.

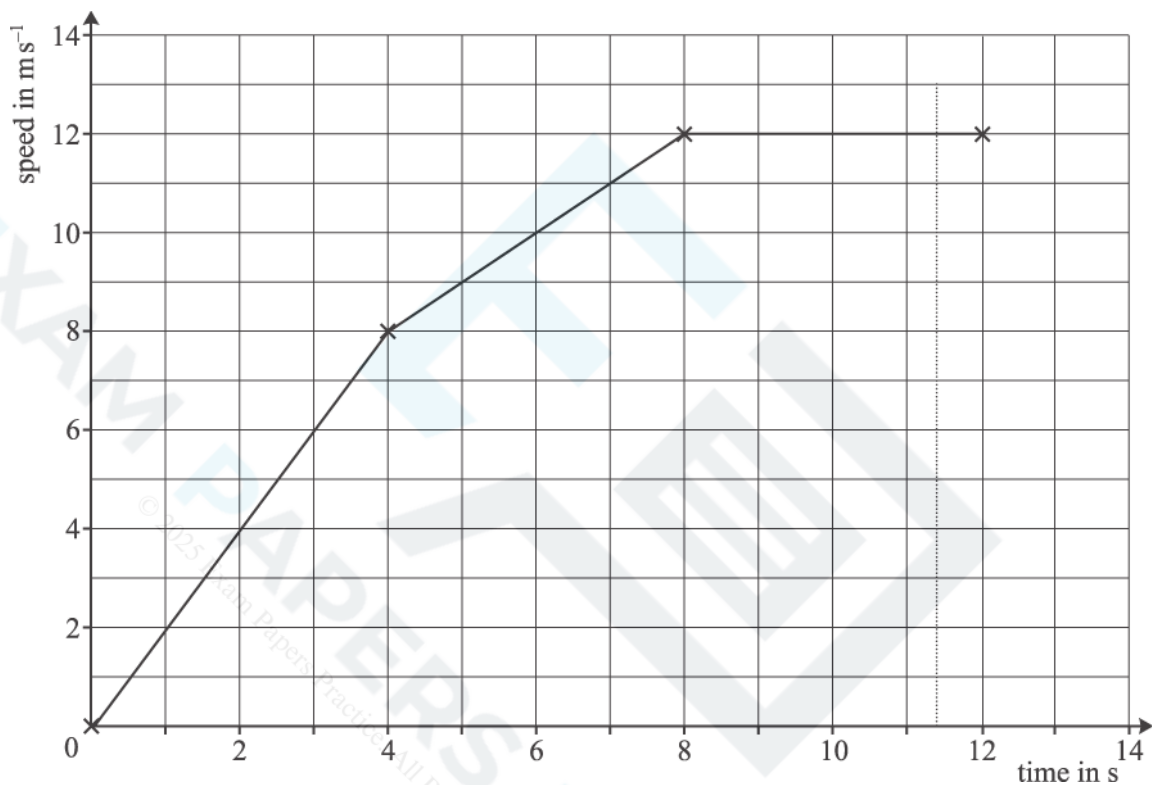


Fig. 6

- (i) Use Model P to estimate

- A the distance the runner has gone at the end of 12 seconds,
- B how long the runner took to complete 100 m.

[6]

A mathematician proposes Model Q in which the runner's speed, $v \text{ ms}^{-1}$ at time $t \text{ s}$, is given by

EXAM PAPERS PRACTICE

$$v = \frac{5}{2}t - \frac{1}{8}t^2.$$

(ii) Verify that Model Q gives the correct speed for $t = 8$.

[1]

(iii) Use Model Q to estimate the distance the runner has gone at the end of 12 seconds.

[4]

(iv) The runner was timed at 11.35 seconds for the 100 m.

Which model places the runner closer to the finishing line at this time?

[3]

(v) Find the greatest acceleration of the runner according to each model.

[4]

- 2 A particle is travelling along a straight line with constant acceleration. P, O and Q are points on the line, as illustrated in Fig. 4. The distance from P to O is 5 m and the distance from O to Q is 30 m.



Fig. 4

Initially the particle is at O. After 10 s, it is at Q and its velocity is 9 ms^{-1} in the direction $\overrightarrow{OQ}$.

(i) Find the initial velocity and the acceleration of the particle.

[4]

(ii) Prove that the particle is never at P.

3 Fig. 3 shows two people, Sam and Tom, pushing a car of mass 1000 kg along a straight line l on level ground.

Sam pushes with a constant horizontal force of 300 N at an angle of 30° to the line l .

Tom pushes with a constant horizontal force of 175 N at an angle of 15° to the line l .

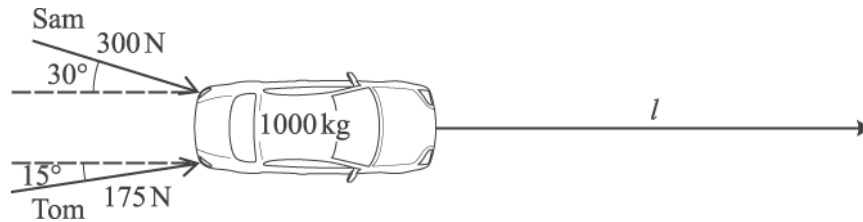


Fig. 3

(i) The car starts at rest and moves with constant acceleration. After 6 seconds it has travelled 7.2 m.

Find its acceleration.

[3]

(ii) Find the resistance force acting on the car along the line l .

[4]

(iii) The resultant of the forces exerted by Sam and Tom is not in the direction of the car's acceleration. Explain briefly why.

[1]

- 4 A particle moves along a straight line through an origin O. Initially the particle is at O.

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At time t s, its displacement from O is x m and its velocity, v ms⁻¹, is given by

$$v = 24 - 18t + 3t^2.$$

- (i) Find the times, T_1 s and T_2 s (where $T_2 > T_1$), at which the particle is stationary.

[2]

- (ii) Find an expression for x at time t s.

Show that when $t = T_1$, $x = 20$ and find the value of x when $t = T_2$.

[4]

- 5 Fig. 5 shows blocks of mass 4 kg and 6 kg on a smooth horizontal table. They are connected by a light, inextensible string. As shown, a horizontal force F N acts on the 4 kg block and a horizontal force of 30 N acts on the 6 kg block.

The magnitude of the acceleration of the system is 2 ms⁻².



Fig.5

- (i) Find the two possible values of F .

[4]

- (ii) Find the tension in the string in each case.

[3]

- 6 The directions of the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are east and north.

The velocity of a particle, v ms⁻¹, at time t s is given by

$$\mathbf{v} = (16 - t^2)\mathbf{i} + (31 - 8t)\mathbf{j}.$$

Find the time at which the particle is travelling on a bearing of 045° and the speed of the particle at this time.

[6]

- 7 Fig. 7 illustrates a train with a locomotive, L, pulling two trucks, A and B.

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The locomotive has mass 90 tonnes and is subject to a resistance force of 2000 N.

Each of the trucks A and B has mass 30 tonnes and is subject to a resistance force of 500 N.

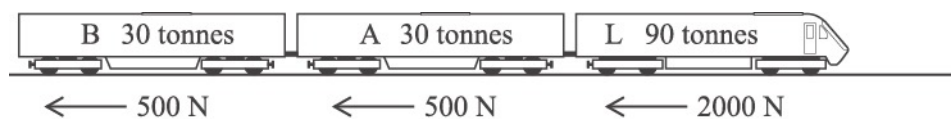


Fig. 7

Initially the train is travelling along a straight horizontal track. The locomotive is exerting a driving force of 12 000 N.

- (i) Find the acceleration of the train.

[3]

- (ii) Find the tension in the coupling between trucks A and B.

[3]

When the train is travelling at 10ms^{-1} , a fault occurs with truck A and the resistance to its motion changes from 500 N to 5000 N.

The driver reduces the driving force to zero and allows the train to slow down under the resistance forces and come to a stop.

- (iii) Find the distance the train travels while slowing down and coming to a stop.

Find also the force in the coupling between trucks A and B while the train is slowing down, and state whether it is a tension or a thrust.

[7]

The fault in truck A is repaired so that the resistance to its motion is again 500 N. The train continues and comes to a place where the track goes up a uniform slope at an angle of α° to the horizontal.

- (iv) When the train is on the slope, it travels at uniform speed. The driving force remains at 12 000 N. Find the value of α .

[3]

- (v) Show that the force in the coupling between trucks A and B has the same value that it had in part (ii).

[2]

- 8 Fig. 1 shows the velocity-time graph of a cyclist travelling along a straight horizontal road between two sets of traffic lights. The velocity, v , is measured in metres per second and the time, t , in seconds. The distance travelled, s metres, is measured from when $t = 0$.

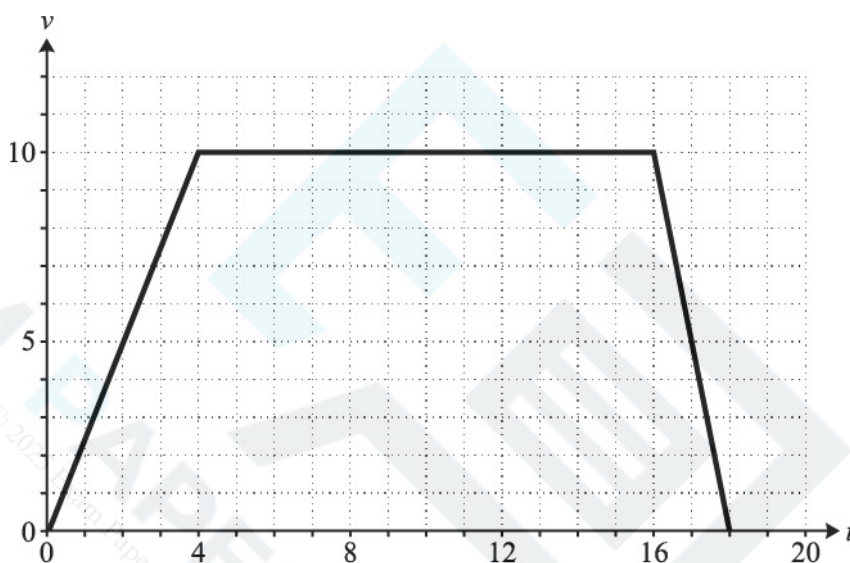


Fig. 1

- (i) Find the values of s when $t = 4$ and when $t = 18$.

[3]

- (ii) Sketch the graph of s against t for $0 \leq t \leq 18$.

[3]

- 9 A box of emergency supplies is dropped to victims of a natural disaster from a stationary helicopter at a height of 1000 metres. The initial velocity of the box is zero.

At time t s after being dropped, the acceleration, $a \text{ m s}^{-2}$, of the box in the vertically downwards direction is modelled by

$$a = 10 - t \quad \text{for} \quad 0 \leq t \leq 10,$$

$$a = 0 \quad \text{for} \quad t > 10.$$

- (i) Find an expression for the velocity, $v \text{ m s}^{-1}$, of the box in the vertically downwards direction in terms of t for $0 \leq t \leq 10$.

Show that for $t > 10$, $v = 50$.

[4]

- (ii) Draw a sketch graph of v against t for $0 \leq t \leq 20$.

[3]

- (iii) Show that the height, $h \text{ m}$, of the box above the ground at time $t \text{ s}$ is given, for $0 \leq t \leq 10$, by

$$h = 1000 - 5t^2 + \frac{1}{6}t^3.$$

Find the height of the box when $t = 10$.

[4]

- (iv) Find the value of t when the box hits the ground.

[2]

- (v) Some of the supplies in the box are damaged when the box hits the ground. So measures are considered to reduce the speed with which the box hits the ground the next time one is dropped. Two different proposals are made. Carry out suitable calculations and then comment on each of them.

A The box should be dropped from a height of 500 m instead of 1000 m.

[2]

B The box should be fitted with a parachute so that its acceleration is given by

$$a = 10 - 2t \quad \text{for} \quad 0 \leq t \leq 5,$$

$$a = 0 \quad \text{for} \quad t > 5.$$

[3]

- 10 The battery on Carol and Martin's car is flat so the car will not start. They hope to be able to "bump start" the car by letting it run down a hill and engaging the engine when the car is going fast enough. Fig. 6.1 shows the road leading away from their house, which is at A. The road is straight, and at all times the car is steered directly along it.

- From A to B the road is horizontal.
- Between B and C, it goes up a hill with a uniform slope of 1.5° to the horizontal.
- Between C and D the road goes down a hill with a uniform slope of 3° to the horizontal. CD is 100 m. (This is the part of the road where they hope to get the car started.)
- From D to E the road is again horizontal.

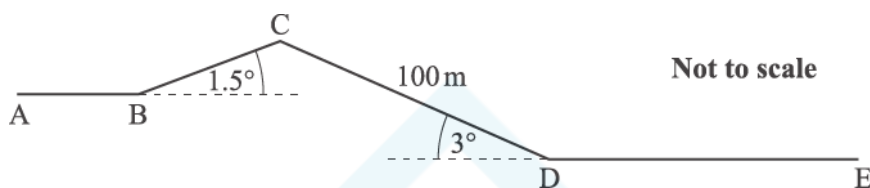


Fig. 6.1

The mass of the car is 750 kg, Carol's mass is 50 kg and Martin's mass is 80 kg.

Throughout the rest of this question, whenever Martin pushes the car, he exerts a force of 300 N along the line of the car.

- (i) Between A and B, Martin pushes the car and Carol sits inside to steer it. The car has an acceleration of 0.25 m s^{-2} .

Show that the resistance to the car's motion is 100 N.

[3]

Throughout the rest of this question you should assume that the resistance to motion is constant at 100 N.

- (ii) They stop at B and then Martin tries to push the car up the hill BC.

Show that Martin cannot push the car up the hill with Carol inside it but can if she gets out.

Find the acceleration of the car when Martin is pushing it and Carol is standing outside.

[6]

- (iii) While between B and C, Carol opens the window of the car and pushes it from outside while steering with one hand. Carol is able to exert a force of 150 N parallel to the surface of the road but at an angle of 30° to the line of the car. This is illustrated in Fig. 6.2.

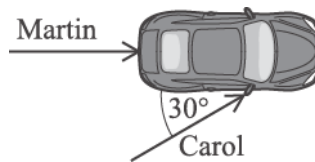


Fig. 6.2

Find the acceleration of the car.

[4]

- (iv) At C, both Martin and Carol get in the car and, starting from rest, let it run down the hill under gravity. If the car reaches a speed of 8 m s^{-1} they can get the engine to start.

Does the car reach this speed before it reaches D?

[5]

- 11 Fig. 4 illustrates a straight horizontal road. A and B are points on the road which are 215 metres apart and M is the mid-point of AB.

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When a car passes A its speed is 12 m s^{-1} in the direction AB. It then accelerates uniformly and when it reaches B its speed is 31 m s^{-1} .

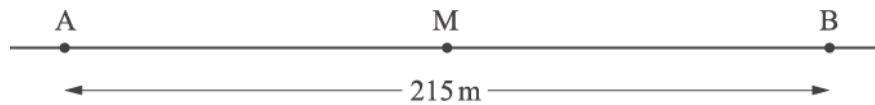


Fig. 4

- (i) Find the car's acceleration. [2]
- (ii) Find how long it takes the car to travel from A to B. [2]
- (iii) Find how long it takes the car to travel from A to M. [3]
- (iv) Explain briefly, in terms of the speed of the car, why the time taken to travel from A to M is more than half the time taken to travel from A to B. [1]

- 12 The map of a large area of open land is marked in 1 km squares and a point near the middle of the area is defined to be the origin. The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are in the directions east and north.

At time t hours the position vectors of two hikers, Ashok and Kumar, are given by:

$$\text{Ashok} \quad \mathbf{r}_A = \begin{pmatrix} -2 \\ 0 \end{pmatrix} + \begin{pmatrix} 8 \\ 1 \end{pmatrix} t,$$

$$\text{Kumar} \quad \mathbf{r}_K = \begin{pmatrix} 7t \\ 10 - 4t \end{pmatrix}.$$

- (i) Prove that the two hikers meet and give the coordinates of the point where this happens.

[4]

- (ii) Compare the speeds of the two hikers.

[3]

- 13 Fig. 7 illustrates a situation on a building site. An unexploded bomb is being lifted by light ropes that pass over smooth pulleys. The ropes are attached to winches V and W.

- The weight of the bomb is 7500 N.
- The winches are on horizontal ground and are at the same level.
- The sloping parts of the ropes from V and W are at angles α and β to the horizontal.
- The point P is level with the horizontal sections of the ropes and is 16 m and 9 m from the two pulleys, as shown.
- The winches are controlled so that the bomb moves in a vertical line through P. The tension in the rope attached to winch W is kept constant at 8000 N. The tension, T N, in the rope attached to winch V is varied.
- The distance between the top of the bomb, B, and the point P is d metres.

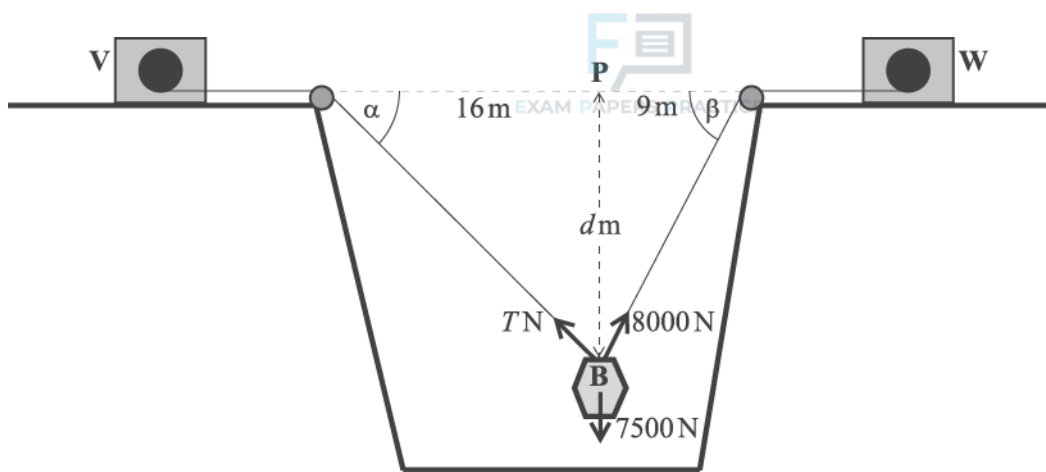


Fig. 7

At a particular stage in the lift, $d = 12$ and $T = 6000$.

- (i) Find the values of $\cos \alpha$ and $\cos \beta$ at this stage.

[1]

- (ii) Verify that, at this stage, the horizontal component of the bomb's acceleration is zero. Find the vertical component of its acceleration.

[7]

At a later stage, the bomb is higher up and so the values of d , T , α and β have all changed.

- (iii) Show that $T = \frac{8000 \cos \beta}{\cos \alpha}$.

Hence show that $T = \frac{4500\sqrt{d^2 + 256}}{\sqrt{d^2 + 81}}$.

[4]

- (iv) Find the acceleration of the bomb when $d = 6.75$.

[4]

- (v) Explain briefly why it is not possible for the bomb to be in equilibrium with B at P.

What could you say about the acceleration of the bomb if B were at P and the tensions in the two ropes were equal?

[2]

- 14 Fig. 1 shows a block of mass M kg being pushed over level ground by means of a light rod. The force, T N, this exerts on the block is along the line of the rod.

The ground is rough.

The rod makes an angle α with the horizontal.

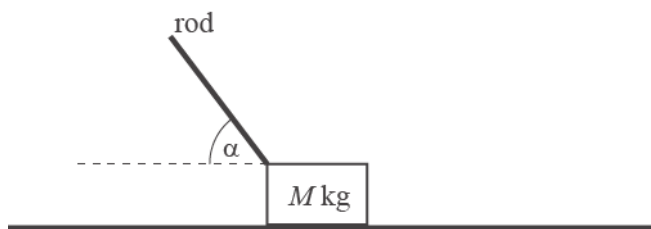


Fig. 1

- (i) Draw a diagram showing all the forces acting on the block.

[3]

- (ii) You are given that $M = 5$, $\alpha = 60^\circ$, $T = 40$ and the acceleration of the block is 1.5 ms^{-2} .

Find the frictional force.

[3]

- 15 A particle P moves on a straight line that contains the point O . At time t seconds the displacement of P from O is s metres, where $s = t^3 - 3t^2 + 3$.

- (a) Determine the times when the particle has zero velocity.

[3]

- (b) Find the distances of P from O at the times when it has zero velocity.

[2]

- 16 A car is usually driven along the whole of a 5 km stretch of road at a constant speed of 25 m s^{-1} . On one occasion, during a period of 50 seconds the speed of the car is as shown by the speed-time graph in Fig. 7; the rest of the 5 km is travelled at 25 m s^{-1} .

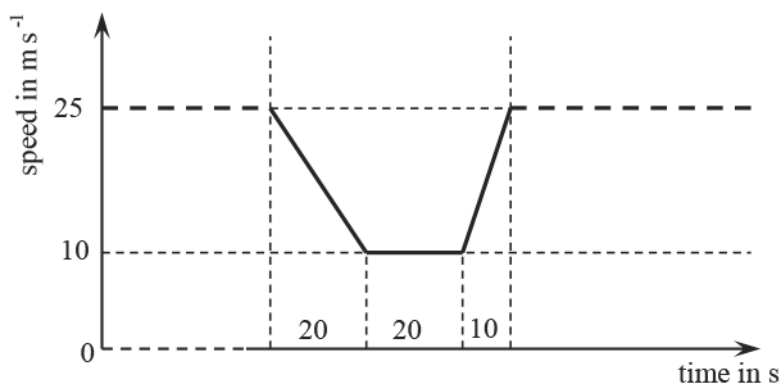


Fig. 7

How much more time than usual did the journey take on this occasion? Show your working clearly.

[4]

- 17 In an experiment a small box is hit across a floor. After it has been hit, the box slides without rotation. The box passes a point A. The distance the box travels after passing A before coming to rest is S metres and the time this takes is T seconds. The only resistance to the box's motion is friction due to the floor.
- The mass of the box is m kg and the frictional force is a constant F N.

(a) (i) Find the equation of motion for the box while it is sliding.

(ii) Show that $S = kT^2$ where $k = \frac{F}{2m}$.

[4]

(b) Given that $k = 1.4$, find the value of the coefficient of friction between the box and the floor.

[4]

- 18 A model boat has velocity $\mathbf{v} = ((2t - 2)\mathbf{i} + (2t + 2)\mathbf{j}) \text{ m s}^{-1}$ for $t \geq 0$, where t is the time in seconds. $\mathbf{i}$ is the unit vector east and $\mathbf{j}$ is the unit vector north. When $t = 3$, the position vector of the boat is $(3\mathbf{i} + 14\mathbf{j}) \text{ m}$.

(a) Show that the boat is never instantaneously at rest. [2]

(b) Determine any times at which the boat is moving directly northwards. [2]

(c) Determine any times at which the boat is north-east of the origin. [5]

19 In this question you must show detailed reasoning.

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A boy plays on a path that runs north-south through an origin O. His displacement x metres north of O at time t seconds is given by

$$x = -0.7t^2 + 4t \text{ for } 0 \leq t \leq 10.$$

(a) Determine the direction in which he is moving when $t = 7$.

[3]

(b) Find the furthest distance from O reached by the boy for $0 \leq t \leq 10$.

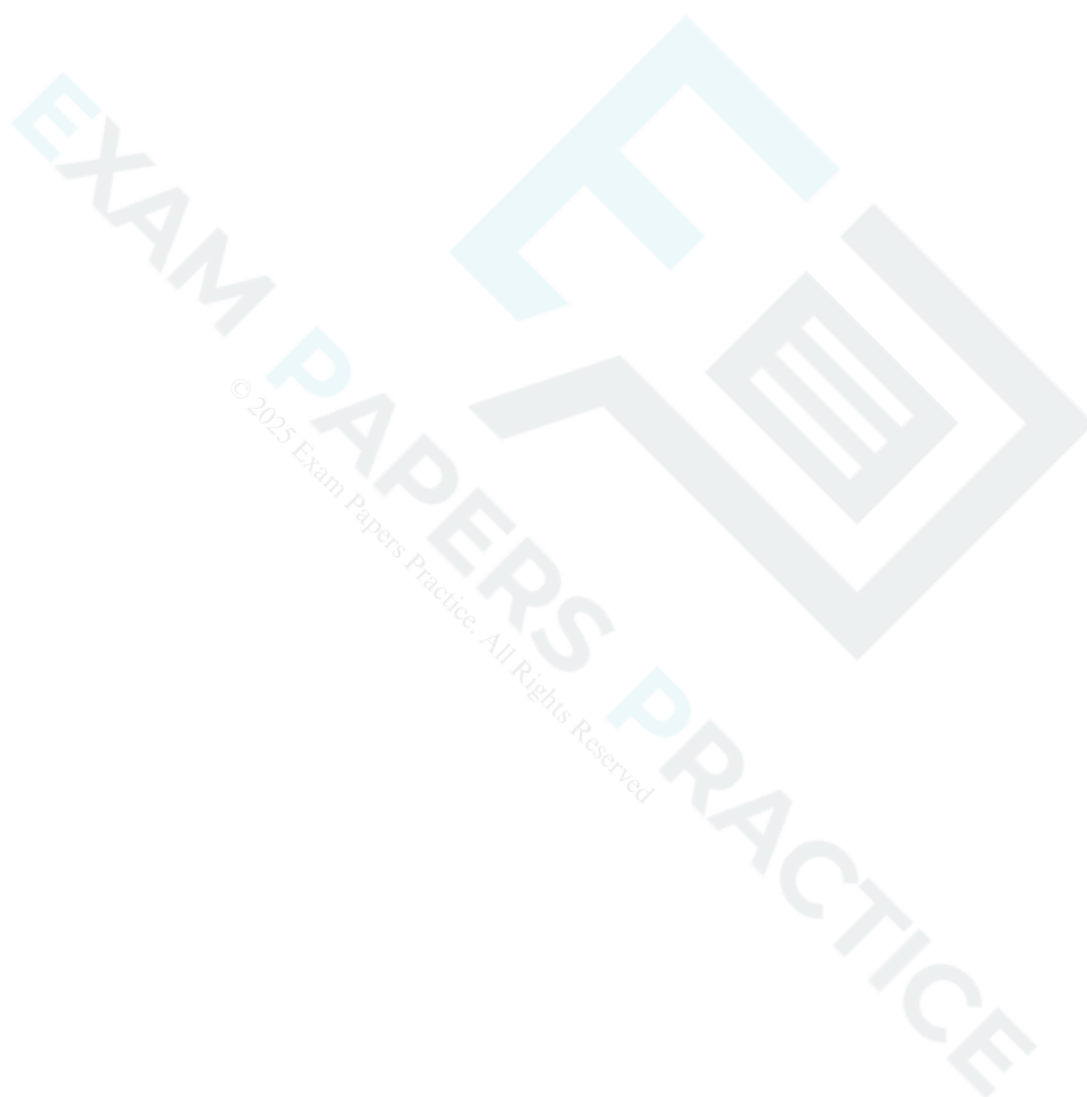
[5]

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- 20 Dorothy and Harold push a wheelbarrow of mass 20 kg. They both push in the direction of motion. The wheelbarrow moves 10m from rest on a straight horizontal path. Dorothy applies a force of 90 N. Harold applies a force of 90 N for the first 3 m, and then reduces the force to 80 N for the remaining 7 m. The resistance to motion is 170 N throughout.

Find the time taken for the wheelbarrow to move the 10 m.

[9]



21 A car travels along a straight track for 5 seconds. Its displacement s metres after t seconds is given by

$$s = 3t + 0.1t^3.$$

Show that the car does not have constant acceleration.

[3]

22 Austin pushes a curling stone of mass 18 kg on ice. He releases the stone and it slides 21 m in a straight line until coming to rest after 14 s. He models the motion by assuming that the frictional force is constant.

(a) Show that the speed of the stone at the moment that Austin releases it is 3 m s^{-1} .

[2]

(b) Find the magnitude of the frictional force acting on the stone.

[3]

(c) Calculate the coefficient of friction between the stone and the ice, giving your answer correct to 2 significant figures.

[3]

- 23 Two cars, A and B, are travelling in different lanes in the same direction along a straight road. The initial situation is illustrated in Fig. 5. EXAM PAPERS PRACTICE

- At this time, A is stationary at traffic lights at O. The lights have just turned green and A is on the point of moving off.
- B is travelling towards O with speed 20 m s^{-1} . B is 75 m behind A.



Fig. 5

During the subsequent motion,

- A has constant acceleration 2 m s^{-2} ,
- the traffic lights remain green and B maintains a constant speed 20 m s^{-1} .

In order to model the subsequent motion you should make two assumptions.

- The cars can overtake each other with no interference from other traffic.
- The position of a car is defined by a point at its front and so the length of the car need not be considered.

- (i) Find the times at which the two cars are side by side.

[4]

(ii) Find the distance A travels while it is behind B.

[2]

(iii) There is a speed camera 400 m from O.
How fast is A travelling when it passes the speed camera?

[2]

- 24 A train is travelling along a straight test track. It starts from rest and reaches its maximum speed after a time of 2 minutes and 21 seconds. During that time it travels 5 km.

Two models, A and B, are considered for its motion.

In Model A, it is assumed that the train has constant acceleration.

- (i) Find the acceleration of the train and its maximum speed according to Model A.

[5]

In Model B, it is assumed that the acceleration, $a \text{ m s}^{-2}$ at time t seconds after starting, is given by

$$a = 0.6 - 3 \times 10^{-5} \times t^2.$$

- (ii) Show that, according to Model B, the time taken for the train to reach its maximum speed is 2 minutes 21.42 seconds (to the nearest 0.01 s).

[2]

- (iii) Find expressions for the speed of the train and the distance that it has travelled at time t , according to Model B.

[4]

- (iv) Hence show that Model B is consistent with the train travelling a distance of 5 km to attain maximum speed.

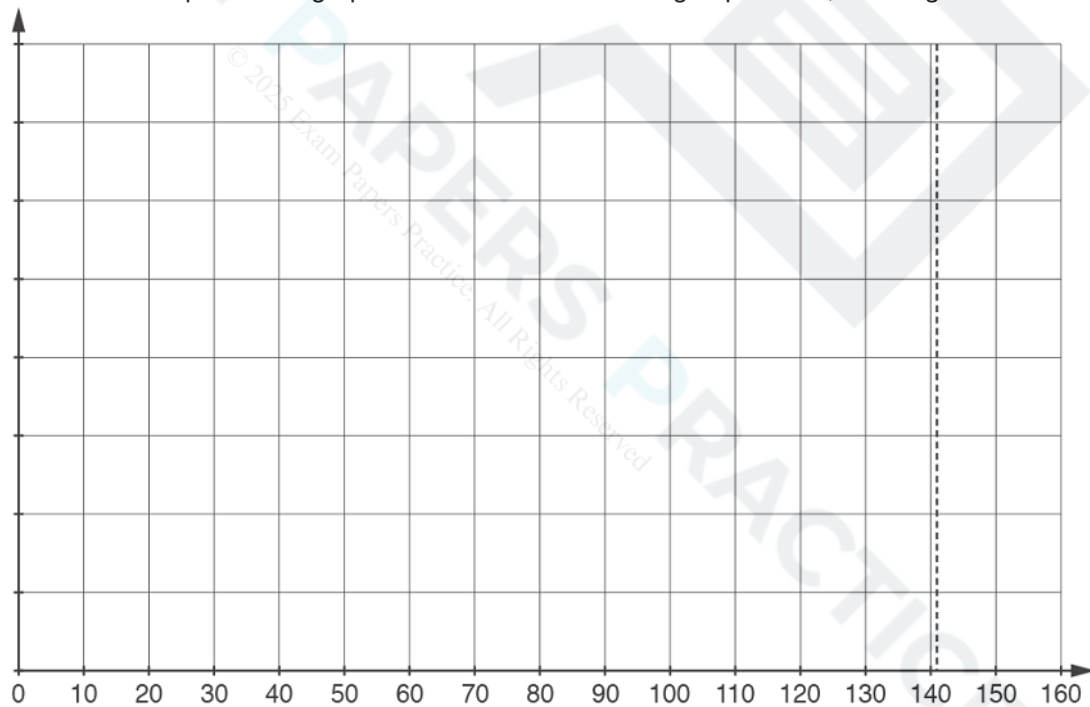
Find the maximum speed of the train according to this model.

[3]

- (v) When the train reaches its maximum speed it continues at that speed.

Draw the speed-time graphs for both models on the grid provided, labelling them A and B.

[4]



- 25 A block slides from rest down a line of greatest slope of a smooth plane inclined at 20° to the horizontal. Calculate the speed of the block when it has travelled 2 m.

[4]

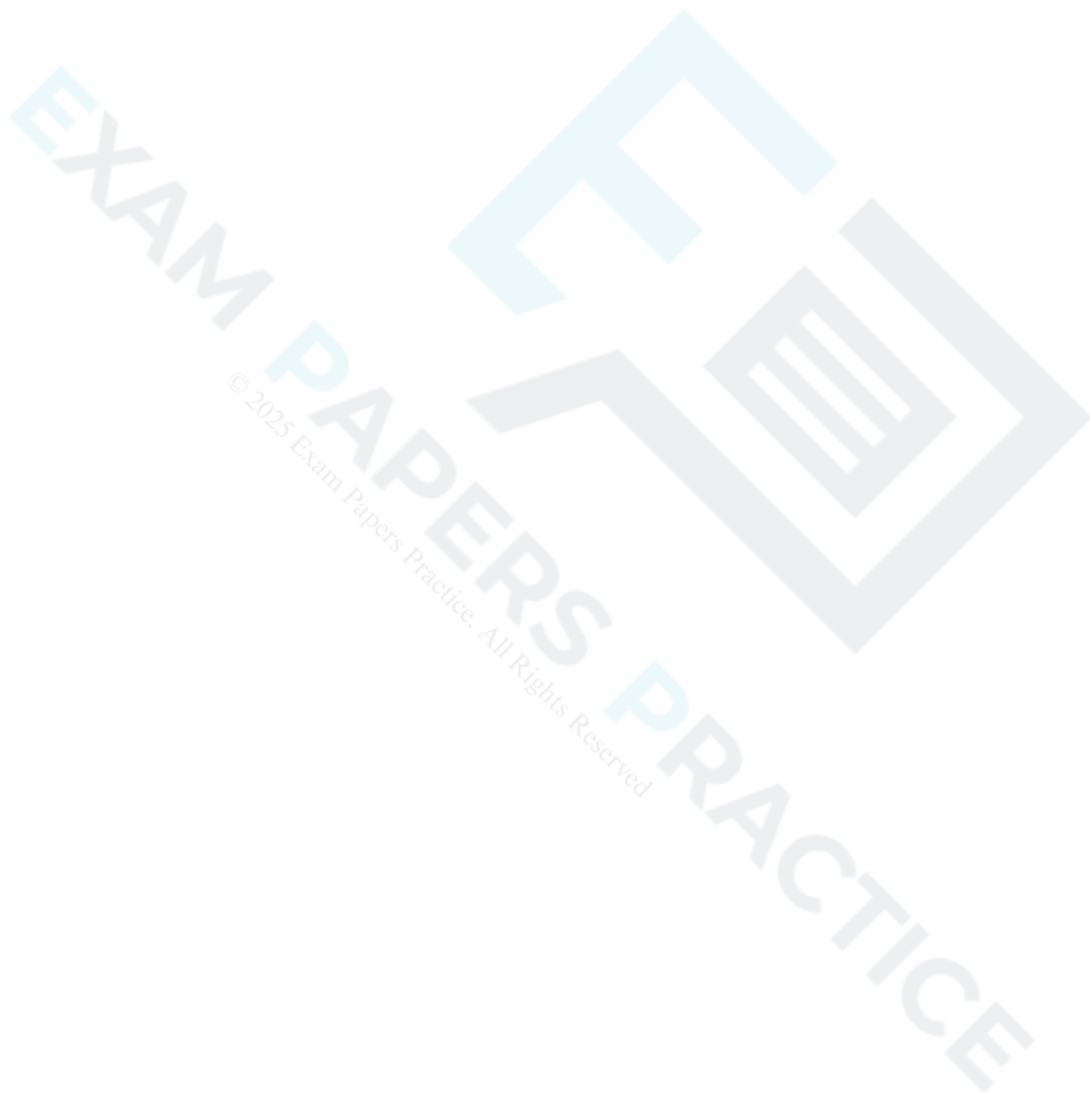


Fig. 6 shows the velocity-time graph for a car as it travels along a straight road. The car sets off from some traffic lights and stops momentarily at a road junction. The velocity $v \text{ ms}^{-1}$ of the car at time $t \text{ s}$ after leaving the traffic lights is modelled by

$$v = 0.025t^3 - 0.8t^2 + 6.4t \text{ for } 0 \leq t \leq 20.$$

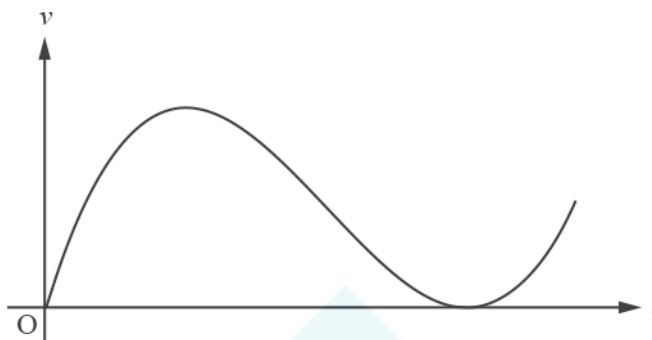


Fig. 6

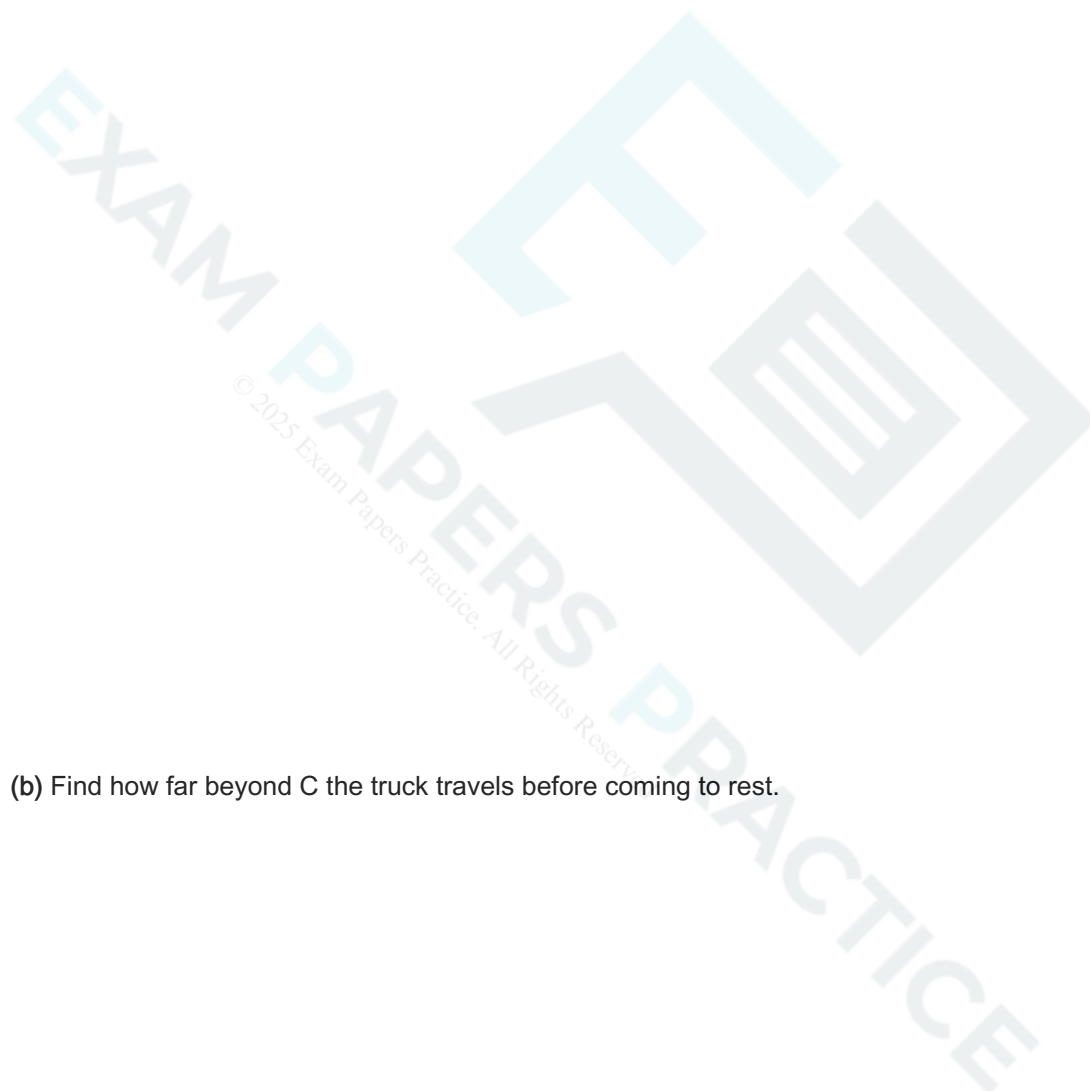
Calculate the distance from the traffic lights to the road junction.

[6]

- 27 A truck is travelling along a straight road ABC, and is slowing down at a constant rate. The truck takes 4 s to travel the 64 m from A to B and it takes another 4 s to travel the 32 m from B to C.

(a) Find

- the speed of the truck at A,
- the acceleration of the truck.



- (b) Find how far beyond C the truck travels before coming to rest.

28 Rory pushes a box of mass 2.8 kg across a rough horizontal floor against a resistance of 19 N . Rory applies a constant horizontal force. The box accelerates from rest to 1.2 ms^{-1} as it travels 1.8 m .

(a) Calculate the acceleration of the box.

[2]

(b) Find the magnitude of the force that Rory applies.

[2]

29 The position vector $\mathbf{r}$ metres of a particle at time t seconds is given by

EXAM PAPERS PRACTICE

$$\mathbf{r} = (1 + 12t - 2t^2) \mathbf{i} + (t^2 - 6t) \mathbf{j}.$$

(a) Find an expression for the velocity of the particle at time t .

[2]

(b) Determine whether the particle is ever stationary.

[2]

- 30 The velocity of a car, $v \text{ ms}^{-1}$ at time t seconds, is being modelled. Initially the car has velocity 5 ms^{-1} and it accelerates to 11.4 ms^{-1} in 4 seconds.

EXAM PAPERS PRACTICE

In model A, the acceleration is assumed to be uniform.

- (a) Find an expression for the velocity of the car at time t using this model.

[3]

- (b) Explain why this model is not appropriate in the long term.

[1]



Model A is refined so that the velocity remains constant once the car reaches 17.8 ms^{-1} .

- (c) Sketch a velocity-time graph for the motion of the car, making clear the time at which the acceleration changes.

[3]

- (d) Calculate the displacement of the car in the first 20 seconds according to this refined model.

[3]



In model B, the velocity of the car is given by

$$v = \begin{cases} 5 + 0.6t^2 - 0.05t^3 & \text{for } 0 \leq t \leq 8, \\ 17.8 & \text{for } 8 < t \leq 20. \end{cases}$$

- (e) Show that this model gives an appropriate value for v when $t = 4$.

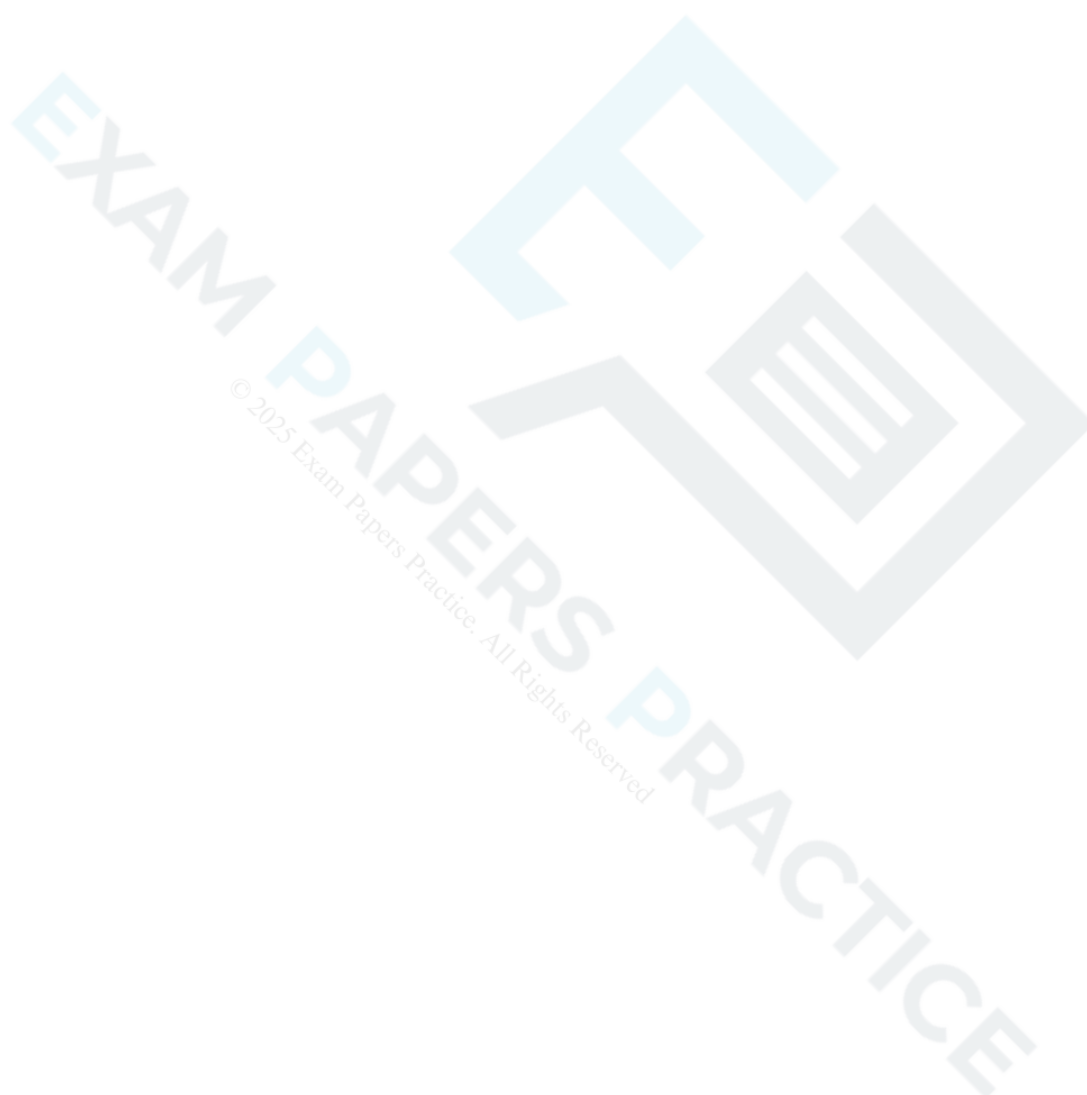
[1]

- (f) Explain why the value of the acceleration immediately before the velocity becomes constant is likely to mean that model B is a better model than model A.

[3]

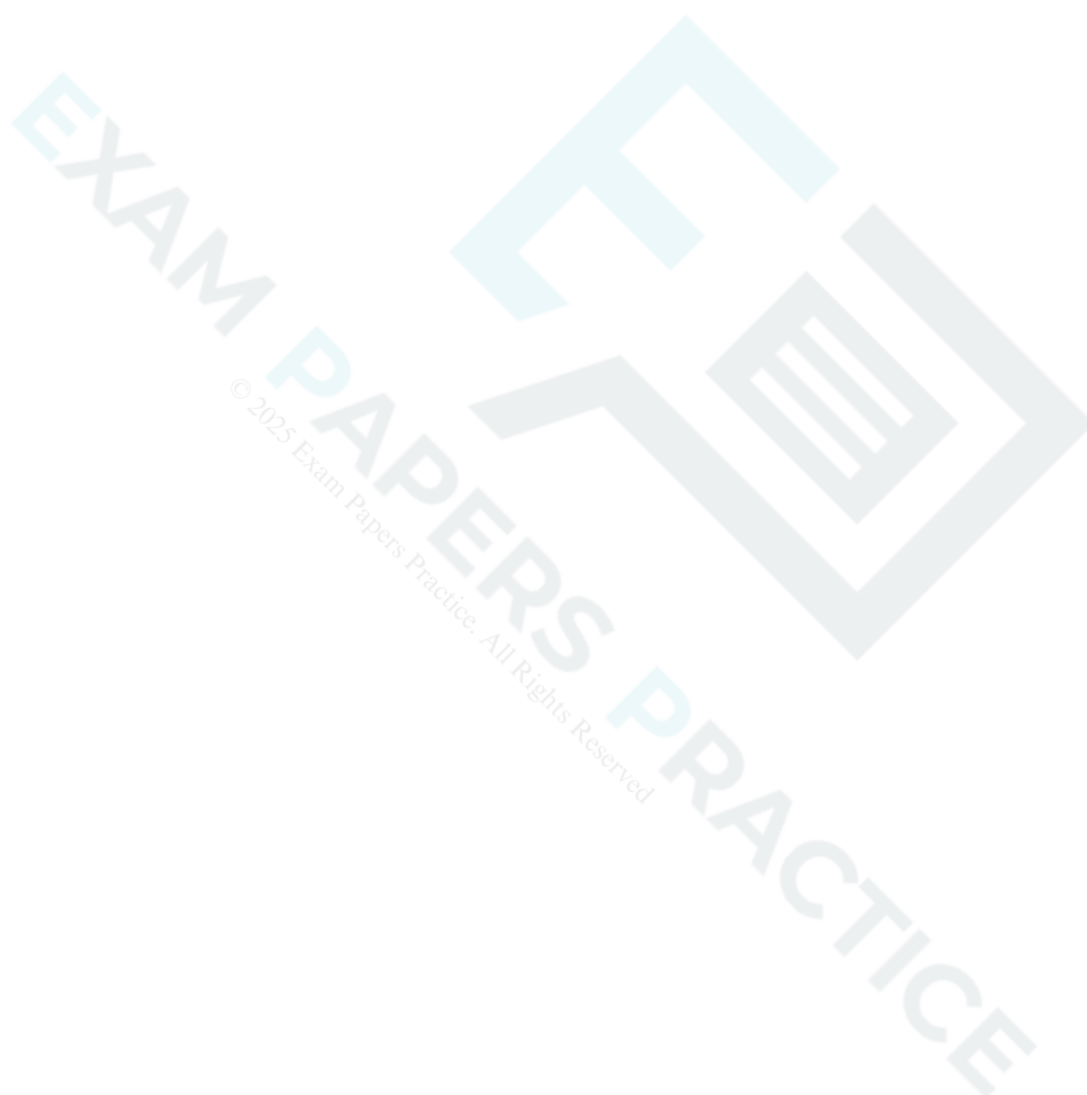
(g) Show that model B gives the same value as model A for the displacement at time 20 s.

[3]



- 31 A toy boat of mass 1.5 kg is pushed across a pond, starting from rest, for 2.5 seconds. During this time, the boat has an acceleration of 2 m s^{-2} . Subsequently, when the only horizontal force acting on the boat is a constant resistance to motion, the boat travels 10 m before coming to rest. Calculate the magnitude of the resistance to motion.

[6]



- 32 Rory runs a distance of 45 m in 12.5 s. He starts from rest and accelerates to a speed of 4 m s^{-1} . He runs the remaining distance at 4 m s^{-1} .

Rory proposes a model in which the acceleration is constant until time T seconds.

- (a) Sketch the velocity-time graph for Rory's run using this model. [2]

- (b) Calculate T . [2]

- (c) Find an expression for Rory's displacement at time t s for $0 \leq t \leq T$. [2]

(d) Use this model to find the time taken for Rory to run the first 4 m.

[1]

Rory proposes a refined model in which the velocity during the acceleration phase is a quadratic function of t . The graph of Rory's quadratic goes through $(0, 0)$ and has its maximum point at $(S, 4)$. In this model the acceleration phase lasts until time S seconds, after which the velocity is constant.

(e) Sketch a velocity-time graph that represents Rory's run using this refined model.

[1]

(f) State with a reason whether S is greater than T or less than T . (You are not required to calculate the value of S .)

[1]

- 33 A bus travelling on a straight road accelerates uniformly from 2.5 m s^{-1} to 7.5 m s^{-1} in 12 s. It then travels at 7.5 m s^{-1} for 20 s before slowing uniformly to rest in 8 s.

(a) Sketch a velocity-time graph for the bus.

[3]

(b) Calculate the average speed of the bus.

[4]

- 34 In a laboratory experiment, the motion of a small object moving in a straight line is being studied. A model for its velocity $v \text{ m s}^{-1}$ at time $t \text{ s}$ is given by

$$v = at^4 + bt^3,$$

where a and b are constants and $t > 0$.

The velocity has maximum value of 0.1 m s^{-1} when $t = 1$.

- (a) Determine the values of a and b .

[5]

(b) Find the time at which the particle changes direction.

[2]

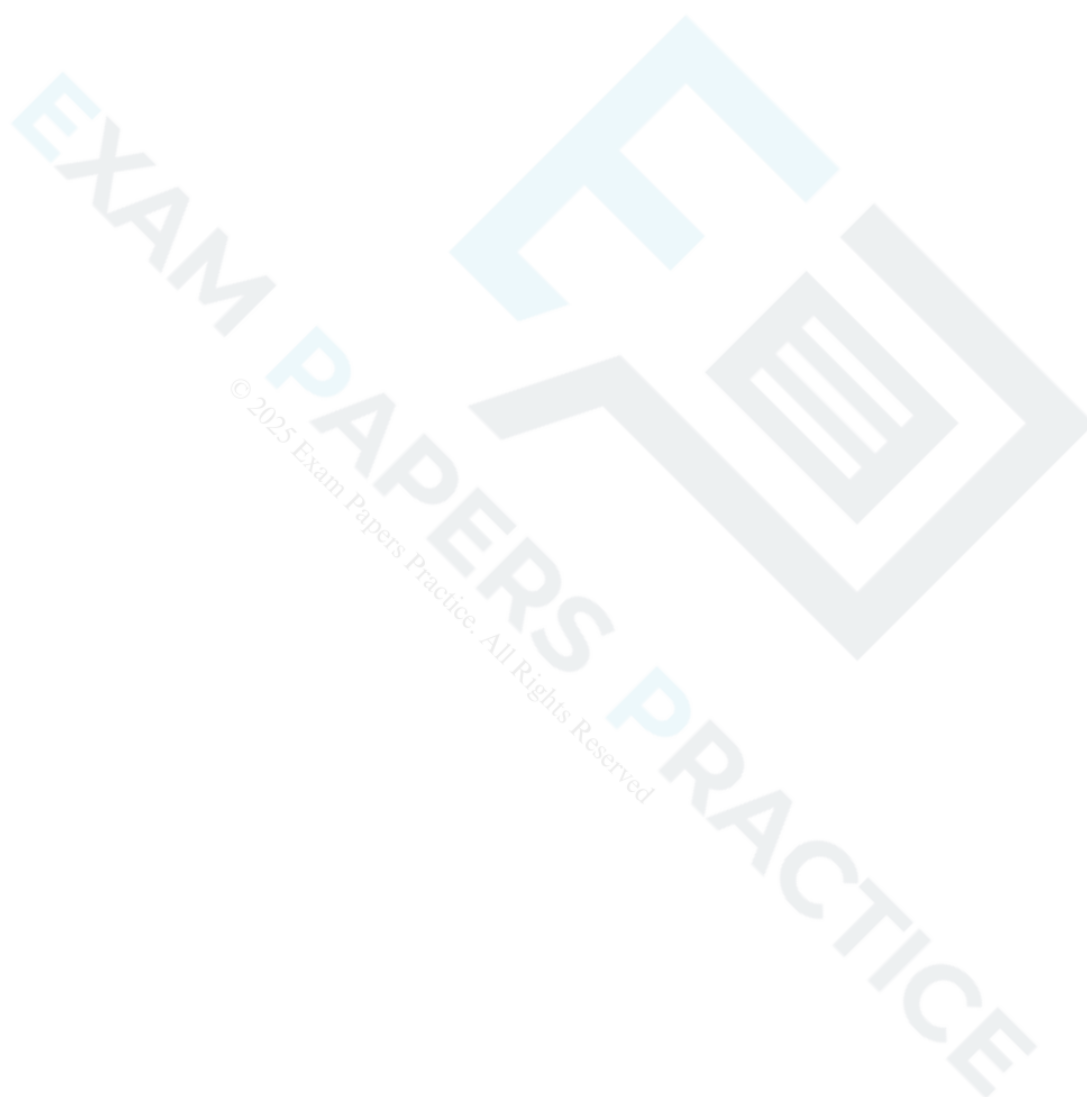
(c) Explain why the model would not be suitable for very large values of t .

[1]

35 A block of mass 2.7 kg is placed on a rough plane inclined at 25° to the horizontal. The coefficient of friction between the block and the plane is 0.4 . A force of 24 N parallel to a line of greatest slope of the plane pulls the block up the plane from rest.

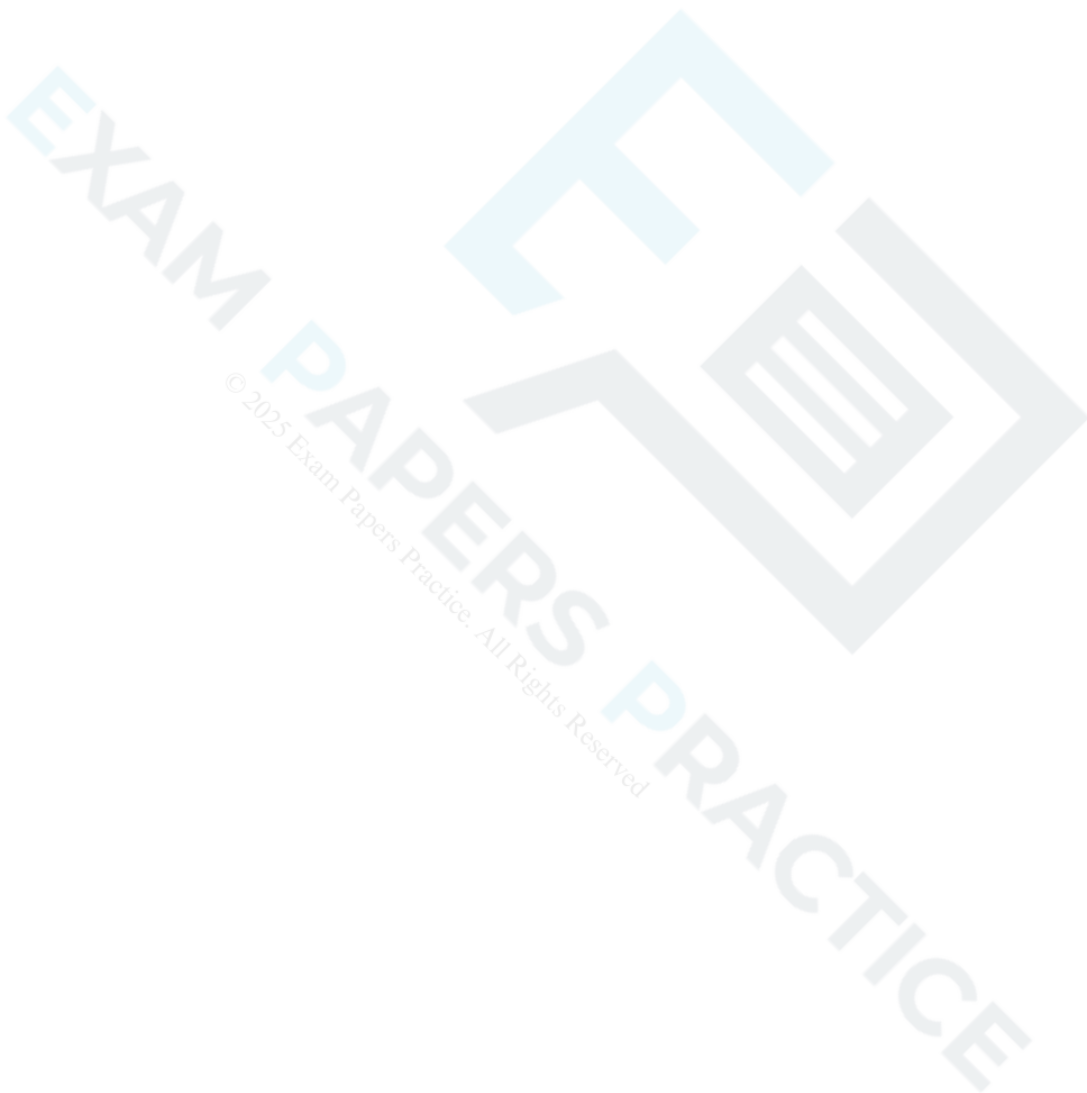
(a) Calculate the acceleration of the block.

[5]



After 5 s the 24 N force is removed.

- (b) Calculate the distance that the block travels after the force is removed before coming instantaneously to rest. [6]



36 In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

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A model boat sails from a point A with an initial velocity of $-2\mathbf{j} \text{ m s}^{-1}$. It accelerates uniformly to a velocity of $(4\mathbf{i} + 6\mathbf{j})\text{ms}^{-1}$ in 8 s.

(a) Calculate the acceleration of the boat.

[2]

(b) Find the time at which the boat is sailing due east.

[2]

4 s after leaving A, the boat is at point B with position vector $(5\mathbf{i} - 2\mathbf{j})\text{m}$.

(c) Find the position vector of A.

[4]

- 37 A cyclist is travelling in a straight line. She has a velocity of 3ms^{-1} when passing O. After 4 s she reaches A which is 24 m from O. After a further 6 s she reaches B which is 80 m beyond A.

Determine whether modelling the motion as having constant acceleration is consistent with these values.

[5]

- 38 Fig. 8 shows the velocity-time graph of a car that is travelling in a straight line as it manoeuvres then drives away. Its velocity ms^{-1} at time t s is given by $v = 0.1t^3 + 0.9t^2 - 1$.

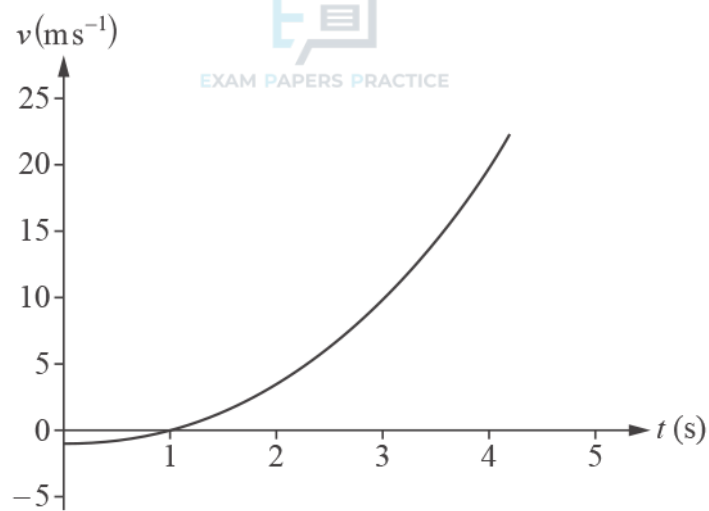


Fig. 8

(a) Describe two features of the motion of the car in the first 4 seconds.

[2]

(b) In this question you must show detailed reasoning.

Calculate the total distance travelled in the first 4 seconds.

[5]

(c) Find an expression for the acceleration of the car in terms of t .

[2]

39 A model train consists of an engine of mass 0.3 kg and a truck of mass 0.2 kg. The train is pulled up a smooth plane inclined at 20° to the horizontal by a force of 2.5 N. This force is applied to the engine and acts parallel to a line of greatest slope of the plane.

(a) Show that the acceleration of the train is 1.65ms^{-2} correct to 3 significant figures. [3]

(b) Find the tension in the coupling between the engine and the truck. [2]

(c) Find the distance travelled by the train as it accelerates from 1ms^{-1} to 3ms^{-1} . [2]

40(a) In this question you must show detailed reasoning.

A car accelerates from rest along a straight level road. The velocity of the car after 8 s is 25.6 m s^{-1} .

In one model for the motion, the velocity $v \text{ m s}^{-1}$ at time t seconds is given by $v = 1.2t^2 - kt^3$, where k is a constant and $0 \leq t \leq 8$.

The model gives the correct velocity of 25.6 m s^{-1} at time 8 s. Show that $k = 0.1$.

[2]

(b) A second model for the motion uses constant acceleration.

Find the value of the acceleration which gives the correct velocity of 25.6 m s^{-1} at time 8 s.

[2]

(c) Show that these two models give the same value for the displacement in the first 8 s.

[5]

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- 41 A car of mass 1200 kg travels from rest along a straight horizontal road. The driving force is 4000 N and the total of all resistances to motion is 800 N.
Calculate the velocity of the car after 9 seconds.

[4]

42 The velocity $v \text{ ms}^{-1}$ of a particle at time $t \text{ s}$ is given by

EXAM PAPERS PRACTICE

$$v = 0.5t(7 - t).$$

Determine whether the **speed** of the particle is increasing or decreasing when $t = 8$.

[4]



43(a) A child is running up and down a path. A simplified model of the child's motion is as follows:

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- he first runs north for 5 s at 4 m s^{-1} ;
- he then suddenly stops and waits for 8 s;
- finally he runs in the opposite direction for 7 s at 3.5 m s^{-1} .

Taking north to be the positive direction, sketch a velocity-time graph for this model of the child's motion.

[2]

(b) Using this model,

calculate the total distance travelled by the child,

[2]

(c) find his final displacement from his original position.

[1]

- 44 A particle is moving in a straight line. The acceleration $a \text{ m s}^{-2}$ of the particle at time $t \text{ s}$ is given by $a = 0.8t + 0.5$. The initial velocity of the particle is 3 m s^{-1} in the positive x -direction.

Determine whether the particle is ever stationary.

[6]



45(a) A block of mass 2 kg is placed on a rough horizontal table. A light inextensible string attached to the block passes over a smooth pulley attached to the edge of the table. The other end of the string is attached to a sphere of mass 0.8 kg which hangs freely.

The part of the string between the block and the pulley is horizontal. The coefficient of friction between the table and the block is 0.35. The system is released from rest.

Draw a force diagram showing all the forces on the block and the sphere.

[3]

(b) Write down the equations of motion for the block and the sphere.

[2]

(c) Show that the acceleration of the system is 0.35 m s^{-2} .

[4]

EXAM PAPERS PRACTICE

(d) Calculate the time for the block to slide the first 0.5 m. Assume the block does not reach the pulley.

[2]

46 A projectile is fired from ground level at 35 m s^{-1} at an angle of θ° above the horizontal.

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State a modelling assumption that is used in the standard projectile model.

[1]



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47(a) In this question you must show detailed reasoning.

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A particle moves in a straight line. Its velocity ms^{-1} after t s is given by $v = t^3 - 5t^2$.

Find the times at which the particle is stationary.

[2]

(b) Find the total distance travelled by the particle in the first 6 seconds.

[3]

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48(a) A car travelling in a straight line accelerates uniformly from rest to $V \text{ ms}^{-1}$ in $T\text{s}$. It then slows down uniformly, coming to rest after a further $2T\text{s}$.

Sketch a velocity-time graph for the car.

[2]

(b) The acceleration in the first stage of the motion is 2.5ms^{-2} and the total distance travelled is 240 m.

Calculate the values of V and T .

[4]

49(a) An astronaut on the surface of the moon drops a ball from a point 2 m above the surface.

EXAM PAPERS PRACTICE

Without any calculations, explain why a standard model using $g = 9.8 \text{ ms}^{-2}$ will not be appropriate to model the fall of the ball.

[1]

(b) The ball takes 1.6 s to hit the surface.

Find the acceleration of the ball which best models its motion. Give your answer correct to 2 significant figures.

[2]

(c) Use this value to predict the maximum height of the ball above the point of projection when thrown vertically upwards with an initial velocity of 15 ms^{-1} .

[2]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1		i	(A) Distance travelled = Area under the graph	M1	Attempt to find area
		i	$\frac{1}{2} \times 4 \times 8 + \frac{1}{2} \times 4 \times (8 + 12) + 4 \times 12$	M1	Splitting into suitable parts
		i	104 m	A1	Cao
					Allow all 3 marks for 104 without any working
		i	(B) Either Working backwards from distance when $t = 12$	M1	
		i	$12 - \frac{(104 - 100)}{12}$	M1	Allow this mark for 0.33... Follow through from their total distance
		i	11.67 s	A1	Cao
		i	Or		
		i	Working forwards from when $t = 8$	M1	
		i	$8 + \frac{(100 - 56)}{12}$	M1	Allow this mark for 3.67... Follow through from their distance at time 8s
		i	11.67 s	A1	Coa
		ii	Substituting $t = 8$ gives $v = \frac{5}{2} \times 8 - \frac{1}{8} \times 8^2 = 12$	B1	
		iii	Distance = $\int_0^{12} \left(\frac{5t}{2} - \frac{t^2}{8} \right) dt$	M1	Integrating v. Condone no limits.
		iii	$\left[\frac{5t^2}{4} - \frac{t^3}{24} \right]_0^{12}$	A1	Condone no limits
		iii	[180-72] (-[0])	M1	Substituting $t = 12$
		iii	108 m	A1	
		iv	Model P: distance at $t = 11.35$ is 96.2	B1	Cao
		iv	Model Q: distance at $t = 11.35$ is		
		iv	$\left[\frac{5t^2}{4} - \frac{t^3}{24} \right]_0^{11.35} = 100.1$	M1	Substituting 11.35 in their expression from part (iii)
		iv	Model Q places the runner closer	E1	Cao from correct previous working for both models

Question			Answer/Indicative content	Marks	Guidance
		v	Model P: Greatest acceleration $\frac{8}{4} = 2 \text{ m s}^{-2}$	B1	Differentiating v Award if correct answer seen Examiner's Comments This was the first of the two Section B questions. It involved two models for the speed of a runner covering 100 metres. It was well answered. In part (i) candidates worked from a given speed-time graph and most were successful in doing so; however some did not realise that the second request needed some calculation and could not be obtained just from reading off the graph. The question then presented the second model as an equation for v in terms of t; most candidates realised that the questions on this involved the use of calculus and answered them correctly. The last two parts of the question involved comparing results from the two models and there were many correct answers, well presented with clear statements as to which model was being considered.
		v	Model Q: $a = \frac{dv}{dt} = \frac{5}{2} - \frac{t}{4}$	M1	
		v		A1	
		v	Model Q: Greatest acceleration is 2.5 ms^{-2}	B1	
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
2		i	Either $s = \frac{1}{2}(u + v)t$ Take O as the origin.	M1	Use of one relevant equation, including substitution
		i	$30 = \frac{1}{2} \times (u + 9) \times 10$		Enter text here.
		i	$u = -3$	A1	
		i	$v = u + at$	M1	Use of a second relevant equation including substitution
		i	$9 = -3 + 10a$		
		i	$a = 1.2$	A1	
		i	or $v = u + at \Rightarrow u + 10a = 9$	M1	Use of one relevant equation, including substitution
		i	$s = ut + \frac{1}{2}at^2 \Rightarrow u + 5a = 3$	M1	Use of a second relevant equation including substitution
		i	Solving simultaneously: $a = 1.2$	A1	
		i	$u = -3$	A1	
		i	or $s = vt - \frac{1}{2}at^2$	M1	Use of one relevant equation, including substitution
		i	$\Rightarrow a = 1.2$	A1	
		i	$v = u + at$	M1	Use of a second relevant equation including substitution
		i	$\Rightarrow u = -3$	A1	Examiner's Comments This question involved a particle travelling under constant acceleration along a straight line, two constant acceleration formulae were required and most candidates obtained the right answers but there were some careless mistakes, and some mistakes in quoting the standard results.
		ii	Either $s = ut + \frac{1}{2}at^2$	3	
		ii	Solving for P: $-5 = -3t + \frac{1}{2} \times 1.2t^2$	M1	Quadratic equation with $s=-5$
		ii	$0.6t^2 - 3t + 5 = 0$		
		ii	Discriminant = $3^2 - 4 \times 0.6 \times 5 = -3$	M1	Considering the discriminant or equivalent

Question			Answer/Indicative content	Marks	Guidance
		ii	No real roots for t ($\Rightarrow$ Particle is never at P)	E1	Cao without wrong working in the whole question.
		ii	Or Find when $v = 0$	M1	
		ii	$v = u + at, v = 0 \Rightarrow t = 2.5$		Or use $v^2 = u^2 + 2as$
		ii	$s = ut + \frac{1}{2}at^2$ and $t = 2.5$	M1	
		ii		E1	Cao without wrong working in the whole question. Comparison necessary
		ii	$\Rightarrow s = -3.75 > -5$		
		ii	Special cases when their $u > 0$ and their $a > 0$	SC1	"It is always going to the right"
		ii		SC1	Demonstration that it is at -5 for two negative times.
					Examiner's Comments This question involved a particle travelling under constant acceleration along a straight line. In part (i) two constant acceleration formulae were required and most candidates obtained the right answers but there were some careless mistakes, and some mistakes in quoting the standard results. In part (ii) candidates were asked to prove that the particle was never at a certain point and it was a pleasure to see how well this was answered, usually either by setting up a quadratic equation and showing it had a negative discriminant, or by finding the turning point in the motion. A handful of candidates just tested some particular cases and no credit was given for this.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
3		i	$s = ut + \frac{1}{2}at^2$	M1	Substitution required
		i	$7.2 = \frac{1}{2} \times a \times 6^2$	A1	
		i	$a = 0.4 \text{ ms}^{-2}$	A1	Coa
		ii	$F = ma$	M1	Attempt at Newton's second law
		ii		M1	Attempt at resolving both S and T
		ii	$300 \cos 30^\circ + 175 \cos 15^\circ - R = 1000 \times 0.4$	A1	(Correct elements present and no extras); follow through for a
		ii	$R=28.8 \text{ N}$	A1	Coa
		iii	The resistance perpendicular to the line of motion has been ignored.	B1	Allow There is also a sideways resistance force Examiner's Comments The first two parts of this question, about two people pushing a car, were well answered. In part (i) almost all candidates found the car's acceleration correctly and in part (ii) many resolved the forces correctly to find the resistance to the car's motion. However, there were few good answers to part (iii); candidates were expected to comment on the sideways resistance to motion (acting on the car's tyres).
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
4		i	$v = 0 \Rightarrow 3(t - 2)(t - 4) = 0$	M1	Setting $v = 0$ (may be implied)
		i	$T_1 = 2, T_2 = 4$	A1	Accept $t = 2$ and $t = 4$
					Examiner's Comments This question was about motion with non-constant acceleration along a straight line. It was very well answered with many candidates obtaining full marks. Candidates used a given equation for v to find when the particle is stationary.
		ii	$x = \int v dt$	M1	Use of integration
		ii	$x = 24t - 9t^2 + t^3 + c : c = 0$	A1	Condone omission of c
		ii	$t = 2 \Rightarrow x = 48 - 36 + 8 = 20$	E1	CAO
		ii	$t = 4 \Rightarrow x = 96 - 144 + 64 = 16$	A1	CAO
					Examiner's Comments This question was about motion with non-constant acceleration along a straight line. It was very well answered with many candidates obtaining full marks. In part (ii) they had to integrate to find an expression for the position and substitute in the two times they had found in part (i). It was pleasing to note an almost complete absence of attempts using constant acceleration formulae.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
5		i	If the acceleration is to the right		
		i	Overall $30 - F = (4 + 6) \times 2$	M1	Newton's 2 nd Law in one direction. No extra forces allowed and signs must be correct.
		i	$F = 10$	A1	
		i	If the acceleration is to the left	M1	For considering second direction. No extra forces allowed and signs must be correct.
		i	$F = 50$	A1	<u>Examiner's Comments</u> This question was about connected particles, in the form of two blocks on a table. Part (i) was best answered treating the system as a whole. Both parts were correctly answered by many candidates. However, a few candidates did not realise that an acceleration of magnitude 2 could be in either direction, to the left or to the right.
		ii	6 kg block $30 - T = 6 \times 2$	M1	Newton's 2 nd law with correct elements on either block
		ii	$\Rightarrow T = 18$	A1	CAO No follow through from part (i)
		ii	In the other case $T = 42$	A1	CAO No follow through from part (i)
					<u>Examiner's Comments</u> This question was about connected particles, in the form of two blocks on a table. Part (ii) asked for the tension in the connecting string and so required candidates to work with one of the blocks. A not uncommon mistake, particularly in part (ii), was to introduce extra forces into the equations of motion.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
6			Equate i and j components of $\mathbf{v}$	M1	The candidate recognises that the i and j components must be equal.
			$16 - t^2 = 31 - 8t$	A1	An equation is formed.
			$t^2 - 8t + 15 = 0$		
			$(t - 3)(t - 5) = 0$		
			$t = 3$ or 5	A1	May be implied by later working.
			When $t = 3$, $\mathbf{v} = 7\mathbf{i} + 7\mathbf{j}$	B1	
			Speed when $t = 3$ is $7\sqrt{2} = 9.9 \text{ m s}^{-1}$	B1	
			The values of the i and j components must both be positive for the bearing to be 045° .	B1	This mark is dependent on obtaining A1 for the result $t = 3$ or 5 . It is awarded if the speed for the case when $t = 5$ is not included (since $t = 5 \Rightarrow \mathbf{v} = -9\mathbf{i} - 9\mathbf{j}$ and the bearing is 225°).
			☐ Alternative trial and error		Note: Candidates who obtain r and equate the east and north components should be awarded SC1 for the whole question.
			The i and j components of $\mathbf{v}$ must be equal	M1	The candidate recognises that the i and j components must be equal.
			The i and j components of $\mathbf{v}$ must both be positive for the bearing to be 045° .	B1	This can be demonstrated during the question either by a suitable convincing diagram including 45° , or by a suitable convincing argument.
			At least one value of t is substituted	A1	Trial and error is used
			$t = 3$	A1	$t = 3$ is found by trial and error
			When $t = 3$, $\mathbf{v} = 7\mathbf{i} + 7\mathbf{j}$	B1	
			Speed when $t = 3$ is $7\sqrt{2} = 9.9 \text{ m s}^{-1}$	B1	
					Note Candidates who obtain r and equate the east and north components should be awarded SC1 for the whole question.

Question			Answer/Indicative content	Marks	Guidance
					Examiner's Comments In this question, candidates were given the velocity of a particle using i, j notation to denote east and north, and they were asked to find when it was travelling on a compass bearing of 045° and its speed at that time. This involved equating the components of $\mathbf{v}$; this gave a quadratic equation, leading to two possible times. Candidates then had to recognise that at one of these times the bearing was 225° not 45°. Many candidates obtained full marks on this question. A few made the mistake of trying to work with position vector instead of the velocity. A common mistake was to fail to eliminate the 225° case. A small number of candidates set out to answer this question using a trial and error method and some credit was given for this.
			Total	6	

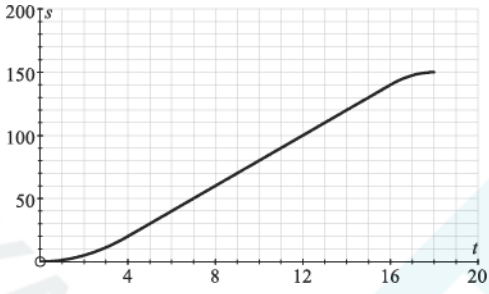
Question			Answer/Indicative content	Marks	Guidance
7		i	Whole train: mass = 150 tonnes	B1	Both totals required. Correct elements must be present CAO. Errors with units (eg not converting tonnes to kilograms) are penalised here but condoned where possible for the remainder of the question. Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Candidates were asked find the acceleration of the whole train and most were successful. A number failed to convert the mass of the train from tonnes into kilograms. This was penalised here but follow-through was then applied for all the marks in the next two parts and for the first two marks in part (iv).
		i	Total Resistance = 3000 N		
		i	$12000 - 3000 = 150000a$	M1	
		i	$a = 0.06$ The acceleration is 0.06 ms^{-2}	A1	
		ii	Truck B: $T - 500 = 30000a$	M1	Correct elements must be present Allow FT for a from part (i) if units are used consistently, for all the marks in this part
		ii	$T - 500 = 30000 \times 0.06$	A1	
		ii	$T = 2300$	A1	
		ii	Between A and B, tension of 2300 N		
		ii	Alternative		

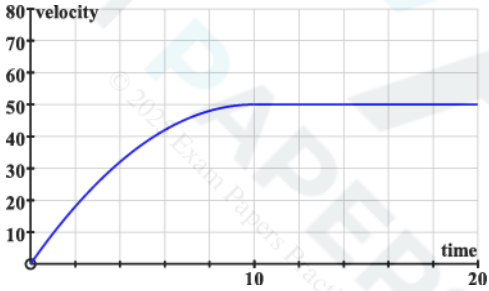
Question			Answer/Indicative content	Marks	Guidance
		ii	Rest of train: $12\,000 - 2500 - T = 120\,000a$	M1	Correct elements must be present Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Candidates were asked to find the tension in the coupling between the two trucks. Most candidates answered this correctly but some introduced extraneous forces.
		ii	$T = 12\,000 - 2500 - 120\,000 \times 0.06$	A1	
		ii	$T = 2300$	A1	
		iii	Treating the train as a whole $-2000 - 5000 - 500 = 150\,000a$	M1	Allow FT for the remaining A marks in part (iii) from an error in a Correct elements must be present. Alternative for rest of train: $-T - 5000 - 2000 = 120\,000 \times -0.05$ The sign of 1000 must be consistent with the direction of T. Dependent on previous M and A marks. Accept "compression".
		iii	$a = -0.05$	A1	
		iii	$v^2 - u^2 = 2as$	M1	
		iii	$0^2 - 10^2 = 2 \times (-0.05) \times s$		
		iii	$s = 1000$ Stopping distance is 1000 m	A1	

Question			Answer/Indicative content	Marks	Guidance
		iii	B: $T - 500 = 30000a$	M1	Correct elements must be present. Alternative for rest of train: $-T - 5000 - 2000 = 120\,000 \times -0.05$
		iii	$T = -1000$	A1	The sign of 1000 must be consistent with the direction of T .
		iii	Between A and B, thrust of 1000 N	A1	Dependent on previous M and A marks. Accept "compression".
					Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Carried 7 marks. The first four of these involved the motion of the train as a whole in a new situation and many candidates obtained all of these marks. The last three marks were for finding the new force in the coupling between the trucks. Most of those candidates who had been successful in part (ii) obtained these marks but others were unable to identify which forces were relevant and which were not.
		iv	Equilibrium parallel to the slope	M1	Correct elements must be present and there must be an attempt to resolve the weight. Condone omission of g for this mark.
		iv	$150000 \times 9.8 \times \sin \alpha + 3000 = 12000$	A1	

Question			Answer/Indicative content	Marks	Guidance
		iv	$\alpha = 0.35^\circ$	A1	CAO Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Involved a new situation in which the train was on a slope. Candidates were asked to find the angle of the slope. While there were plenty of correct answers, many of them were not very well explained. Good force diagrams were something of a rarity.
		v	B: $T_2 - 500 - 30000 \times 9.8 \times \sin 0.35\dots^\circ = 0$ © 2025 Exam Papers Practice. All Rights Reserved	M1	Correct elements must be present. Condone omission of g for this mark. Do not accept 1800 N for the component of the weight without justification. Alternative for rest of train: $12\,000 = T + 2500 + 120\,000 \times 9.8 \times \sin 0.35^\circ$

Question			Answer/Indicative content	Marks	Guidance
		v	$T_2 = 2300$ Between A and B, tension of 2300 N, as in part (ii)	A1	This mark can only be awarded if the angle found in (iv) is correct. Examiner's Comments This question was a good source of marks for many candidates. It was about the motion of a train and the forces in one of the couplings. Most candidates were able to answer the parts that involved considering the train as a whole. Fewer were successful when it came to working with part of the train to find the force in a particular coupling. Provided the last two marks on the paper. It exemplified the interesting (and little known) result that the tensions between trucks when the train is going up a slope at constant speed are the same as those when it is accelerating on level ground, under the same driving force. In this part candidates were asked to do no more than find the same numerical answer as they had obtained in part (ii). Stronger candidates were successful on this but many others had failed to find the correct earlier results (the tension in part (ii) and the angle in part (iv)) that were needed here.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
8		i	When $t = 4$, $s = \frac{1}{2} \times 4 \times 10$		Finding the area of the triangle or equivalent.
		i	$s = 20$	B1	
		i	When $t = 18$, $s = \frac{1}{2} \times (18 + 12) \times 10$	M1	A complete method of finding the area of the trapezium or equivalent.
		i	$s = 150$	A1	CAO
		ii			
		ii	Graph joining (0,0), (4,20) and (18, 150)	B1	Allow FT for their (4,20) and (18, 150) Condone extension to (20, 150) with a horizontal line.
		ii	The graph goes through (16, 140)	B1	Allow SC1 for the first two marks if there is a consistent displacement from a correct scale, eg plotting (18,150) at (19, 150)
		ii	Curves at both ends	B1	The sections from $t = 0$ to $t = 4$ and from $t = 16$ to $t = 18$ are both curves Examiner's Comments This question, about interpreting a velocity-time graph, was well answered. It ended with a request to draw the equivalent distance-time graph, parts of which were curves; many candidates did not realise this and so lost the final mark.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
9		i	Integrate a to obtain v	M1	Attempt to integrate Substitution of $t = 10$ to find v Sound argument required for given answer. It must in some way refer to $a = 0$. Examiner's Comments Candidates were asked to find the velocity in each stage of the motion. This required the use of calculus and was quite well-answered. A common mistake was to omit the explanation that the terminal velocity is constant because the acceleration is known to be zero.
		i	$v = 10t - \frac{1}{2}t^2 \quad (+c)$	A1	
		i	$t = 10 \Rightarrow v = 100 - 50 = 50$	M1	
		i	Since $a = 0$ for $t > 10$, $v = 50$ for $t > 10$	A1	
		ii	Continuous two part $v-t$ graph 	B1	The graph must cover $t = 0$ to $t = 20$ B1 B1 B0 if no vertical scale is given Examiner's Comments Involved drawing a velocity-time graph for the two stages of the motion. This was well-answered but some candidates gave a straight line rather than a curve for the first stage.
		ii	Curve for $0 \leq t \leq 10$	B1	
		ii	Horizontal straight line for $10 \leq t \leq 20$	B1	
		iii	Distance fallen = $\int \left(10t - \frac{1}{2}t^2\right) dt$	M1	Attempt to integrate A1
		iii	$d = 5t^2 - \frac{1}{6}t^3 + c \quad (c = 0)$	A1	
		iii	Height = $1000 - d$		

Question			Answer/Indicative content	Marks	Guidance
		iii	Height = $1000 - 5t^2 + \frac{1}{6}t^3$	A1	This mark should only be given if the signs are correctly obtained. oe Examiner's Comments This part involved going from the velocity of the box to its height and so involved further use of calculus. That part was well-done but many candidates failed to deal adequately with the sign change involved in going from the distance fallen from the helicopter to the height above the ground. <u>Marking guidance</u> Follow through between parts of this question should be allowed for the value of h (when $t = 10$) found in part (iii) if it is used in part (iv) or in part (v)(A).
		iii	When $t = 10$, $h = 667$	B1	
		iv	Time at constant vel = $667 \div 50 = 13.3$	B1	FT for h from part (iii) FT Examiner's Comments Candidates were asked for the time taken for the box to fall. Only a few candidates got this right. Almost all forgot that there were two stages to the motion. Many set the cubic expression for height equal to zero and then tried to solve the resulting cubic equation. The correct approach was nothing like as difficult as that.
		iv	Total time $t = 10 + 13.3 = 23.3$	B1	
		v	(A) Since $500 > 333$	M1	For finding the height at which the crate reaches terminal velocity, eg $h = 167$, or equivalent relevant calculation. FT for h from part (iii) if used. Allow either one (or both) of these two statements. Integration to find v
		v	The box will have reached terminal speed. So there is no improvement	A1	
		v	(B) $v = 10t - t^2$ (for $t \leq 5$)	M1	
		v	Terminal velocity is 25 ms^{-1}	A1	

Question			Answer/Indicative content	Marks	Guidance
		v	So better	A1	Examiner's Comments Two alternative methods of trying to limit the damage to the box on landing were considered. The first, in part (A), involved reducing the height from which the box was dropped; however, it would still have reached terminal velocity so this would not have helped. This was not answered very well with many candidates missing the point. The second, in part (B), involved fitting a parachute and this was very well answered, with many candidates obtaining full marks on it. Overall this question was not answered badly. It was not difficult but a high mark on it did require clear thinking.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
10		i	$F - R = ma$	M1	Use of Newton's 2 nd Law
		i	$300 - R = (750 + 50) \times 0.25$	A1	Correct elements present
		i	$R = 100$	A1	This is a given result
					Examiner's Comments Generally this question was answered very well and many candidates got it completely right. Almost all candidates were successful on this part involving the use of Newton's 2 nd Law. Marking guidance Follow through between parts of this question should be allowed for values found in parts (ii) and (iii) providing the questions are not simplified.
		ii	Carol in Component of weight down slope	M1	Resolving down the slope. Accept use of 750 instead of 800.
		ii	$= 800g \sin 1.5^\circ (= 205.2 \text{ N})$	A1	For this mark only condone no g and allow sin-cos interchange.
		ii	Martin has to overcome 305.2 N		
		ii	$300 < 305.2$ Martin cannot manage	A1	Give M1 A1 for $800g \sin 15^\circ$ seen
		ii	Carol out Martin has to overcome $750g \sin 1.5^\circ + 100 = 292.4 \text{ N}$		This mark may be awarded for an argument based on Newton's 2 nd law leading towards $a = -0.006$
		ii	$300 > 292.4$ so Martin manages	B1	Explanation, based on correct working, that Martin can manage. This can be given retrospectively with a comment on a positive value for a .
		ii	$300 - 292.4 = 7.6 = 750a$	M1	Use of Newton's 2 nd Law

Question			Answer/Indicative content	Marks	Guidance
		ii	The acceleration is 0.010 ms^{-2}	A1	Cao. Accept 0.01 or an answer that rounds to 0.01. Examiner's Comments Was about pushing a car up a slope. This was well-answered with most candidates resolving the weight correctly to find its component down the slope. Very few made the mistake of interchanging sin and cos.
		iii	Component of Carol's force parallel to the line of the car	M1	For attempt at resolution in the correct direction. For this mark only, condone sin-cos interchange.
		iii	$= 150\cos 30^\circ (= 129.9)$	A1	Give M1 A1 for $150\cos 30^\circ$ seen
		iii	Resultant forward force = $7.6 + 129.9 = 137.5$ $750a = 137.5$	M1	All forces parallel to the slope present and correct. Sign errors condoned.
		iii	The acceleration is 0.183 ms^{-2}	A1	FT their force parallel to the slope from part (ii) (correct value 7.6 N) Examiner's Comments The car was still being pushed up the slope but an extra force was now involved. This meant that the equation of motion had five terms and some candidates lost marks by omitting one (or more) of them.
		iv	Component of weight down the slope $= (750 + 50 + 80) \times 9.8 \times \sin 3^\circ$		
		iv	$880a = 451.3 - 100$	M1	Newton's 2nd law with correct elements present. No sin-cos interchange. The same mass must be used in both places.
		iv	$a = 0.399$	A1	
			$v^2 - u^2 = 2as$		
		iv	When $v = 8$, $s = 8^2 \div (2 \times 0.399)$	M1	Selection and use of an appropriate formula (unless with $a = g$)
		iv	$s = 80.1$	A1	FT their value of a

Question			Answer/Indicative content	Marks	Guidance
		iv	80.1 < 100 so Yes they get the car started	A1	FT their value of a
		iv	Alternative: Finding the speed after 100 m		
			Component of weight down the slope		
		iv	$= (750 + 50 + 80) \times 9.8 \times \sin 3^\circ$	M1	Newton's 2nd law with correct elements present. No sin-cos interchange
		iv	$880a = 451.3 - 100$		
		iv	$a = 0.399$	A1	
			$v^2 - u^2 = 2as$		
		iv	$v^2 = (0^2) + 2 \times 0.399 \times 100$	M1	Selection and use of an appropriate formula (unless with $a = g$)
		iv	$v = (\sqrt{79.8}) = 8.93...$	A1	FT their value of a
		iv	$(v > 8)$ so they get the car started	A1	FT their value of a
					Examiner's Comments
					The car travelled down a slope and many candidates were able to select the right information and methods to deal with this new situation. Some lost marks by forgetting one of the terms, for example the resistance.
					Overall the response to this question was very pleasing.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
11		i	$v^2 - u^2 = 2as$		
		i	$31^2 - 12^2 = 2 \times 215 \times a$	M1	Selection and use of appropriate equation(s)
		i	$a = 1.9$ so 1.9 ms^{-2}	A1	
		ii	$v = u + at$		
		ii	$31 = 12 + 1.9t$	M1	Selection and use of appropriate equation(s)
		ii	$t = 10$ so 10s	A1	FT from their value of a from part (i).
		iii	$s = ut + \frac{1}{2}at^2$		
		iii	$\frac{215}{2} = 12t + \frac{1}{2} \times 1.9 \times t^2$	M1	Selection and use of $s = ut + \frac{1}{2}at^2$, oe.
		iii	$\left(t = \frac{-12 \pm \sqrt{12^2 + 4 \times 0.95 \times 107.5}}{1.9} \right)$	M1	Correct elements but condone minor arithmetic errors.
		iii	$t = 6.055$ (or -18.69)	A1	Use of quadratic formula (may be implied by answer), oe.
		iii	Alternative: Finding a 2-stage method		FT their a only.
		iii	$v^2 - u^2 = 2as$ and $s = \frac{(u+v)}{2}t$		
		iii	$v = \pm \sqrt{12^2 + 2 \times 1.9 \times 107.5} = (\pm)23.505...$	M1	Selection and use of a complete valid 2-stage method
		iii	$s = \frac{(u+v)}{2}t \Rightarrow t = \frac{2 \times 107.5}{(12 + 23.505...)} \left(\text{or } t = \frac{2 \times 107.5}{(12 - 23.505...)} \right)$	M1	Using the output from the first stage to find t
		iii	$t = 6.055$ (or 18.69)	A1	FT their a only.

Question			Answer/Indicative content	Marks	Guidance
		iv	Because it is accelerating, it travels less fast in the first half of the distance and so takes more time.	B1	The answer must refer to the two parts of the distance (or “the same distance”) so no credit is given to answers like “Because it is accelerating” and “Because its speed is not uniform”. Most successful answers will refer to the times to cover AM and MB but this may be implicit. So B1 should be given for an answer like “It is travelling faster between M and B than it is between A and M” Notice that the fact that the acceleration is uniform is irrelevant. Examiner's Comments This question, about the motion of a car on a road, was very well answered and nearly all candidates got the first 7 marks for selecting and using appropriate constant acceleration formulae. The final part, for 1 mark, asked them to explain why the time to the mid-point of the distance is more than half the time for the whole journey (the car was accelerating throughout). This produced a number of excellent answers but many were unconvincing. A good answer made the point that the distance is the same in both parts but, because it is accelerating, the car is travelling slower in the first part than in the second part, and so takes more time.
			Total	8	

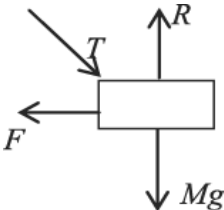
Question			Answer/Indicative content	Marks	Guidance
12		i	Either $-2 + 8t = 7t$ Or $t = 10 - 4t$	M1	Forming an equation for t . Accept vector equation for this mark. May be implied by a statement that $t = 2$.
		i	$\Rightarrow t = 2$	A1	
		i	Substituting $t = 2$ in both expressions	B1	oe, eg showing $t = 2$ satisfies both equations or a vector equation.
		i	They meet at (14, 2)	B1	Accept $\begin{pmatrix} 14 \\ 2 \end{pmatrix}$ Examiner's Comments Candidates were asked to prove the two hikers meet and this involved showing their position vectors are the same at some time ($t = 2$). Many lost a mark by not showing that this was true for both components when the position vectors were equated.
		ii	Ashok's speed is $\sqrt{8^2 + 1^2} = \sqrt{65}$	B1	
		ii	Kumar's speed is $\sqrt{7^2 + (-4)^2} = \sqrt{65} \text{ km h}^{-1}$	B1	
		ii	They both walk at the same speed	B1	CAO from correct speeds SC1 for finding both velocities correctly but neither speed Examiner's Comments Candidates were asked to compare the speeds of the two hikers and most got this right. A few compared their velocities. Others did not know how to obtain the velocities from the given expressions in terms of t for the position vectors.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
13		i	$\cos \alpha = 0.8, \cos \beta = 0.6.$	B1	Or equivalent statements Examiner's Comments This was the second of the long questions on the paper, worth 18 marks. It was set in the context of raising an unexploded bomb from a hole on a building site. This question was quite challenging and many candidates were unsuccessful on the later parts. The question started with a straightforward piece of trigonometry for 1 mark, and almost all candidates were successful. Answers 0.8 and 0.6.
		ii	Horizontal forces $\rightarrow 8000\cos\beta - 6000\cos\alpha$	M1	Do not allow sin-cos interchange
		ii	$4800 - 4800 = 0$ So the horizontal component of acceleration is 0	A1	Must state acceleration is zero
		ii	Vertical forces $\uparrow T\sin\alpha + 8000 \sin \beta - 7500$	M1	Do not allow if the weight is missing Allow $T\cos \beta + 8000 \cos \alpha - 7500$
		ii	$\sin \alpha (= \cos \beta) = 0.6$ and $\sin \beta (= \cos \alpha) = 0.8$	B1	o.e. CAO May be seen or implied in the working
		ii	$6400 + 3600 - 7500 = 2500$	A1	CAO
		ii	Mass of bomb $\frac{7500}{9.8}$ (=765.3) kg	M1	

Question			Answer/Indicative content	Marks	Guidance
		ii	$a = \frac{2500}{765.3} = 3.27$ The acceleration is 3.27 ms^{-1} upwards	A1	CAO Allow 3.26 Examiner's Comments The question then went on to consider the horizontal and vertical components of acceleration in a particular situation. The first demand was to show that the horizontal component is zero. Most candidates got this right but some lost a mark by failing to take the step of going from zero resultant force to zero acceleration; this was a given result and so a high standard of argument was expected. The question then went on to find the vertical component of acceleration and this elicited many good answers. A few candidates failed to convert the weight of the bomb to its mass, and some missed it out completely. There were fewer sin-cos interchanges than might have been the case a few years ago. Answer 3.27 ms^{-2}.
		iii	No horizontal acceleration $\square$ Resultant = 0		
		iii	Horizontal forces $\rightarrow 8000\cos\beta - T\cos\alpha = 0$	M1	Horizontal must be indicated
		iii	$T = \frac{8000\cos\beta}{\cos\alpha}$	A1	
		iii	$\cos\alpha = \frac{16}{\sqrt{d^2 + 16^2}}, \cos\beta = \frac{9}{\sqrt{d^2 + 9^2}}$	M1	

Question			Answer/Indicative content	Marks	Guidance
		iii	$T = 8000 \times \frac{9}{\frac{\sqrt{d^2 + 81}}{16}} = \frac{4500\sqrt{d^2 + 256}}{\sqrt{d^2 + 81}}$	A1	Examiner's Comments The question then went on to consider a general situation during the lift. The first request was derive a given result for T. Many candidates lost marks here by not relating it to the horizontal direction. Some candidates may not have been aware that because this was a given result a high standard of argument was expected. Candidates were then asked to show that the given result for T could be written in a different form. While there were plenty of correct answers, there were also many that appeared to conjure the given result out of incorrect working.
		iv	When $d = 6.75$, $T = 4500 \times \frac{\sqrt{6.75^2 + 256}}{\sqrt{6.75^2 + 81}}$ (= 6946.2 ...)	B1	May be implied by subsequent working
		iv	Vertical forces $\uparrow 6946.2\sin\alpha + 8000\sin\beta - 7500$	M1	Note In this situation $\alpha = 22.9^\circ$, $\beta = 36.9^\circ$ Their α and β . No sin-cos interchange.
		iv	= 0	A1	Note The forces are 2700 N and 4800N Condone any resultant force that rounds to 0 to the nearest integer.
		iv	So the (vertical) acceleration is zero.	A1	CAO Examiner's Comments In this part of the question, the bomb was at the height at which its vertical acceleration was zero and candidates were expected to use the result given at the end of part (iii) to discover this. Only the stronger candidates were successful. Many of those who attempted to find the equation of motion used the wrong angles or the wrong tensions, or forgot about the weight completely. Answer Acceleration = 0 ms⁻²
		iv	Alternative		
		iv	Vertical forces $\uparrow T \sin\alpha + 8000\sin\beta - 7500$ $4500 \times \frac{\sqrt{6.75^2 + 256}}{\sqrt{6.75^2 + 81}} \times \frac{6.75}{\sqrt{6.75^2 + 256}} + 8000 \times \frac{6.75}{\sqrt{6.75^2 + 81}} - 7500$	M1	

Question			Answer/Indicative content	Marks	Guidance
		iv		B1	
		iv	$12500 \times \frac{6.75}{11.25} - 7500 = 0$	A1	
		iv	So the (vertical) acceleration is zero.	A1	CAO
		v	When at P there would be no vertical components of the tensions to counteract the weight.	B1	
		v	The acceleration would be g vertically downwards.	B1	The acceleration must be stated to be g Examiner's Comments In part (ii) the bomb was in a position where it was accelerating upwards. In part (iv) it was in a position where equilibrium was possible but there could be no upwards acceleration. The final part of the question considered the hypothetical situation where the bomb was at the top and so was at the same level as the winches. Candidates were asked to explain why equilibrium was impossible in this situation and to state the acceleration. While there were some excellent explanations many were garbled or wrong. Many candidates said there would be zero acceleration. Answer $g \text{ ms}^{-2}$ vertically downwards.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
14		i		B1	Forces B0 if one force missing or an extra force present
		i		B1	Labels
		i		B1	Arrows B0 if T in tension Allow T given in components provided it is clear they are not additional forces. Allow sin-cos interchange in this case. Give B0 B0 B0 if 2 or more forces missing Examiner's Comments In this question candidates were asked to draw a force diagram and many did this successfully. The most common mistake was to reverse the direction the thrust applied to the block, making it into a tension instead; a few candidates omitted the normal reaction. Some candidates replaced the thrust by its vertical and horizontal components and that was entirely acceptable provided that they were not presented as extra forces in addition to the actual thrust.
		ii			Notice that the same solution applies if the direction of T was wrong in part (i), and full marks are available for part (ii) in this case.
		ii	$T \cos \alpha - F = ma$	M1	Horizontal equation of motion with the right 3 elements
		ii	$40 \cos \alpha - F = 5 \times 1.5$	A1	A0 if sin-cos interchange

Question			Answer/Indicative content	Marks	Guidance
		ii	$F = 12.5$ Frictional force of 12.5 N.	A1	CAO Examiner's Comments In part (ii) candidates were expected to apply Newton's 2nd law to the block and to deduce the frictional force acting on it. Most candidates got this right but there were a few sign errors. Answer 12.5 N
			Total	6	
15		a	Velocity v is $\frac{ds}{dt} = 3t^2 - 6t$ $= 0$ $t = 0$ or 2	M1(AO1.1a) M1(AO1.1) A1(AO1.1) [3]	Attempt to find $\frac{ds}{dt}$ $\frac{ds}{dt} = 0$ must be stated Both roots found
		b	$s(0) = 3$ so distance 3 m $s(2) = 8 - 12 + 3 = -1$ so distance is 1 m	A1(AO1.1) A1(AO3.4) [2]	Accept seeing 3 without comment -1 for s must be seen as well as 1 m for distance
			Total	5	

Question		Answer/Indicative content	Marks	Guidance	
16		Find how much less distance travelled in the 50 s Distance is the area (of trapezium and is) $\frac{(25-10) \times (50+20)}{2} = 525 \text{ m}$ This distance is made up at 25 m s^{-1} to give extra time Extra time is $\frac{525}{25} = 21$ Alternative method Find the distance travelled in the 50 s Find the time for the rest of the journey + 50 and subtract $\frac{5000}{25} = 200$ Distance travelled in the 50 s is 725 m Extra time is $\frac{(5000-725)}{25} + 50 - 200 = 21$	M1(AO3.1b) A1(AO1.1) M1(AO3.4) A1(AO3.2a) M1(AO3.1b) M1(AO3.4) A1(AO1.1) A1(AO3.2a) [4]	Sensible attempt at method including finding distance as an area cao. Need not be evaluated. Many correct routes. FT their area Sensible attempt at method including finding distance as an area May be scored later. oe cao. Many correct routes to find area FT their area. Many correct routes here. Award full marks for 21 seen www	
		Total	4		

Question			Answer/Indicative content	Marks	Guidance
17		a	A Let acceleration be a in the direction of motion. N2L in direction of motion gives $-F = ma$ so $a = -\frac{F}{m}$, which is constant.	M1(AO3.3) A1(AO1.1)	Decide to use N2L to find acceleration No need to say 'constant'
		a	B (As a constant) use <i>suvat</i>, giving $S = 0 \times T - \frac{1}{2} \times \left(-\frac{F}{m}\right) T^2$ so $S = \left(\frac{F}{2m}\right) T^2$, and $k = \frac{F}{2m}$	B1(AO2.1) E1(AO2.4) [4]	Use appropriate (sequence of) <i>Suvat</i>
		b	As sliding, friction is limiting and $F = \mu R$ $R = mg$ $k = \frac{F}{2m}$ so $k = \frac{\mu mg}{2m}$ Hence $\mu = \frac{2k}{g} = \frac{2 \times 1.4}{9.8} = \frac{2}{7}$	M1(AO3.3) A1(AO3.4) M1(AO1.1) A1(AO2.2a) [4]	In $F = \mu R$, substitute for F & R in terms of m and g Or 0.286 (3s.f.)
Total				8	

Question			Answer/Indicative content	Marks	Guidance
18		a	Require both components zero at the same <i>time</i> i component zero only when $t = 1$ and j component only when $t = -1$ so there are no such times	M1(AO3.1b) A1(AO2.4) [2]	May be implied but must be clear Or say j component ≥ 2 since $t \geq 0$
		b	This requires use of the velocity vector Travelling due north means that the i component is zero and the j component +ve So we need $2t - 2 = 0$ for i component, giving $t = 1$. This gives j component $4 > 0$ so yes at $t = 1$.	M1(AO3.3) A1(AO2.4) [2]	Recognise velocity vector required Must test j component
		c	This requires use of the position vector either $\mathbf{r} = \int \mathbf{v} dt$ so $\mathbf{r} = \int ((2t - 2)\mathbf{i} + (2t + 2)\mathbf{j}) dt = (t^2 - 2t + C)\mathbf{i} + (t^2 + 2t + D)\mathbf{j}$ $\mathbf{r} = 3\mathbf{i} + 14\mathbf{j}$ when $t = 3$ so $C = 0$ and $D = -1$ so $\mathbf{r} = (t^2 - 2t)\mathbf{i} + (t^2 + 2t - 1)\mathbf{j}$ Or $\mathbf{a} = 2\mathbf{i} + 2\mathbf{j}$ when $t = 3$ $\mathbf{v} = 4\mathbf{i} + 8\mathbf{j}$, $\mathbf{r} = (4\mathbf{i} + 8\mathbf{j})(t - 3) + \frac{1}{2}(2\mathbf{i} + 2\mathbf{j})(t - 3)^2 + 3\mathbf{i} + 14\mathbf{j}$ and so $\mathbf{r} = (t^2 - 2t)\mathbf{i} + (t^2 + 2t - 1)\mathbf{j}$ the boat is NE of O when the i and j components are equal and +ve we require $t^2 - 2t = t^2 + 2t - 1$ so $t = 0.25$ this gives components of -0.4375 so no.	M1(AO3.1b) M1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO1.1) M1(AO3.2b) A1(AO2.1) [5]	Recognise position vector required May use + C instead Must find a but may omit $3\mathbf{i} + 14\mathbf{j}$ Award even if +ve not mentioned Must be complete argument
			Total	9	

Question			Answer/Indicative content	Marks	Guidance	
19		a	DR $v = -0.7 \times 2t + 4$ When $t = 7$, $v = -5.8$ Boy is moving due south since v is negative	M1(AO 3.1b) M1(AO 3.4) E1(AO 2.2a) [3]	For attempt to differentiate For substitution in their v Dependent on correct -5.8	
		b	DR Max/min s when $v = 0$, so $-1.4t + 4 = 0$ $t = \frac{20}{7}$ When $t = \frac{20}{7}$, $s = \frac{40}{7} = 5.71$ to 3sf When $t = 10$, $s = -30$ and when $t = 0$, $s = 0$ Greatest distance is 30 m	M1(AO 3.1b) A1(AO 1.1b) A1FT(AO 1.1b) B1(AO 1.1b) B1FT(AO 3.2a) [5]	For attempt to solve their $v = 0$ Follow through their t For checking both end-points Must select (their) correct distance, must be positive and must have units	
			Total	8		

Question		Answer/Indicative content	Marks	Guidance
20		$90 + 90 - 170 = 20a$ $a = 0.5$ $3 = 0 + \frac{1}{2} \times 0.5t^2$ $t = \sqrt{12}$ $v = 0 + 0.5\sqrt{12}$ $= \sqrt{3}$ In second phase resultant force is zero so $a = 0$ Time at constant speed $= \frac{7}{\sqrt{3}}$ $\text{Total time} = 2\sqrt{3} + \frac{7\sqrt{3}}{3} = 3.464... + 4.041... = 7.51 \text{ s}$ (3sf)	M1(AO 3.3) A1(AO 1.1b) M1(AO 1.1a) A1(AO 1.1b) M1(AO 3.4) A1(AO 1.1b) B1(AO 3.3) M1(AO 1.1a) A1(AO 1.1b) [9]	For use of Newton II For use of <i>suvat</i> with $s = 3$ and $u = 0$ 3.464... For use of <i>suvat</i> with their a and t to find v 1.732... Allow for 'constant speed' seen
		Total	9	

Question			Answer/Indicative content	Marks	Guidance
21			$v = \frac{ds}{dt} = 3 + 0.3t^2 \text{ and } a = \frac{dv}{dt} = 0.3 \times 2t$ $a = 0.6t$ Acceleration is a function of t so is not constant Alternative method For constant acceleration, $s = ut + \frac{1}{2}at^2 (+c)$ i.e. s is a quadratic function of t So this cubic function is not constant acceleration	M1(AO 1.1a) A1(AO 2.1) E1(AO 2.2a) M1 A1 E1 [3]	Attempt to differentiate twice Must be clearly argued Comparing with standard formula Explicit identification of quadratic, oe Deduction clearly stated
			Total	3	

Question			Answer/Indicative content	Marks	Guidance	
22		a	$21 = \frac{1}{2}(u+0)14$ $u = 3 \text{ AG}$	M1(AO 1.1a) A1(AO 2.1) [2]	Use of <i>suvat</i> equation(s) to find u	
		b	$0 = 3^2 + 2a \times 21$ $a = -\frac{3}{14}$ Magnitude of force = $ ma = 18 \times \frac{3}{14}$ $= \frac{27}{7} = 3.86 \text{ N}$	M1(AO 3.1b) M1(AO 1.1a) A1(AO 1.1b) [3]	Use of <i>suvat</i> to find acceleration Use of Newton II Must be magnitude (positive)	
		c	$N = 18g$ $3.86 = \mu \times 18g$ $\mu = 0.022 \text{ to } 2 \text{ sf}$	B1(AO 3.1b) M1(AO 3.4) A1(AO 1.1b) [3]	soi Use of $F = \mu N$ to find μ Must be given correct to 2sf	
			Total	8		

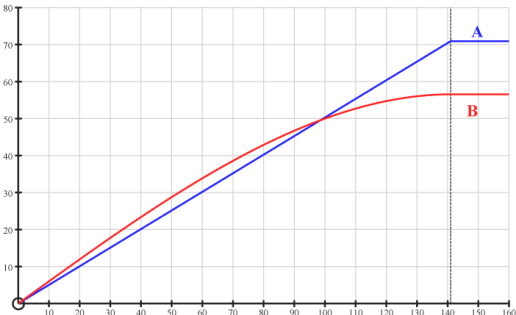
Question			Answer/Indicative content	Marks	Guidance
23		i	A: $x = t^2$ B: $x = -75 + 20t$ When the cars are side by side, $t^2 = -75 + 20t$ $t^2 - 20t + 75 = 0$ $(t - 5)(t - 15) = 0$ The times are 5 seconds and 15 seconds	B1 B1 M1 A1 [4]	Or displacements from B's start point: $x = t^2 + 75$ and $x = 20t$ Must be consistent For equating two distances even if inconsistent Examiner's Comments This question involved the motion of two cars in parallel lanes on a road. One had constant acceleration and the other travelled at constant speed. One car was initially behind the other. In part (i) candidates were asked to find the times when the cars were side by side. While there were many fully correct answers to this, there were also plenty of sign errors involving the 75 m difference in starting position. A few candidates did not realise that answering this question involved setting up an equation for the time t.

Question			Answer/Indicative content	Marks	Guidance
		ii	For A, $s = 225$ when $t = 15$ $s = 25$ when $t = 5$ So A is behind B for 200 m Alternative Using motion of B Speed of B is (constant at) 20 m s^{-1} So between $t = 5$ and $t = 15$, B travels $20 \times (15 - 5) (= 200 \text{ m})$ So A is behind B for 200 m Alternative Using motion of A with the clock re-set When the cars are first level, the motion of A is defined by $u = 10$ and $a = 2$. If the clock is re-set at this moment, $t = 0$ In this case, when they are next level, $t = 10$ $s = ut + \frac{1}{2}at^2 \Rightarrow s = 10 \times 10 + \frac{1}{2} \times 2 \times 10^2$ $\Rightarrow s = 200$	M1 A1 [2] M1 A1 M1 A1	FT for two positive times from part (i) for the M mark only Both values of s attempted CAO Or equivalent, eg using $s = ut + \frac{1}{2}at^2$ with $a = 0$ Or $v = u + at \Rightarrow v = 10 + 2 \times 10 = 30$ followed by use of $v^2 - u^2 = 2as$ Examiner's Comments In part (ii) the question for the distance for which one particular car was ahead of the other. This was lower scoring than the other parts of the question.

Question			Answer/Indicative content	Marks	Guidance
		iii	For A $v^2 - u^2 = 2as$ $v^2 = 2 \times 2 \times 400$ $v = 40$, (so speed 40 ms^{-1}). Alternative finding the time first For A $s = 400 \Rightarrow t = 20$ $v = u + at$ $\Rightarrow v = 40$ so 40 m s^{-1}	M1 A1 [2] M1 A1	There must be an attempt to use the formula There must be evidence of a complete method that can lead to the value of v Examiner's Comments In part (iii) they were asked to find the speed of one of the cars after it had travelled 400 m and this was very well answered, even by those who had made mistakes on earlier parts.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
24		i	2 minutes 21 seconds is 141 seconds $s = ut + \frac{1}{2}at^2$ $5000 = 0 + 0.5 \times a \times 141^2$ $a = 0.503 \text{ (ms}^{-2}\text{)}$ $v = u + at$ $v = 0.503 \times 141 = 70.9 \text{ so } 70.9 \text{ ms}^{-1}$ Alternative using $s = \frac{1}{2}(u + v)t$ $5000 = \frac{1}{2} \times (0 + v) \times 141$ $v = \frac{10000}{141} = 70.9... \text{ so } 70.9 \text{ ms}^{-1}$	B1 M1 A1 M1 A1 [5] M1 A1	Allow 0.50 but not 0.5 Or equivalent, eg $v^2 - u^2 = 2as$ CAO (including 70.5 m s⁻¹) CAO Examiner's Comments This was the first of the two long questions. It was based on two different models for the motion of a train from rest to maximum speed. It involved both constant and variable acceleration. On the whole this question was well answered with many high marks. Part (i) was based on a constant acceleration model. It was very well answered with most candidates obtaining all the five available marks. However, many candidates lost one mark by giving the acceleration to only one significant figure.

Question			Answer/Indicative content	Marks	Guidance
		iv	Substituting $t = 141.42\dots$ in $s = 0.3t^2 - 0.000\ 0025t^4$ $s = 5000$ so consistent with 5 km Substituting $t = 141.42\dots$ in $v = 0.6t - 0.000\ 01t^3$ $v = 56.57\text{ ms}^{-1}$	M1 A1 B1 [3]	Allow substituting $s = 5000$ to show that $t = 141.42\dots$ Notice that $141.42\dots = \sqrt{20\ 000}$ and so the answer of 5000 is exact Examiner's Comments In part (iv) candidates were expected to use the time given in part (ii) and their expression for the distance travelled from part (iii) to verify the distance the train had travelled in attaining maximum speed. Many knew just what to do and were successful. Some tried to do the question in reverse, substituting the distance and forming a quartic equation for the time; this was a viable approach and a few candidates realised that their equation could be written as a quadratic in t^2 and went on to solve it. The question continued to ask for the maximum speed and this was well answered, even by those who had not been successful with the distance.

Question			Answer/Indicative content	Marks	Guidance
		v		B1 B1 B1 B1 [4]	Vertical scale from 0 that ensures that at least half the height is used. For the remaining marks do not allow FT from earlier parts. A: A straight line from (0, 0) to (141, 70.9) B: A curve from (0, 0) to (141, 56.6); the curvature must be in the right sense. Both A from (141, 70.9) to (160, 70.9) and B from (141, 56.6) to (160, 56.6). CAO. Examiner's Comments The final part (v) required the two models to be shown on a speed-time graph. This produced a wide spread of marks. Most candidates knew what they were trying to do but made errors. Some lost a mark by not showing the motion after the train had reached maximum speed and many others drew a straight line rather than a curve for the variable acceleration model.
			Total	18	
25			Resolve down the slope, using Newtons' second law $mg \sin 20^\circ = ma$ $v^2 = 0^2 + 2(g \sin 20^\circ)s$ $v = \sqrt{13.407...} = 3.66 \text{ m s}^{-1}$	M1(AO3.3) A1(AO1.1b) M1(AO1.1a) A1(AO1.1b) [4]	Use of $v^2 = u^2 + 2as$
			Total	4	

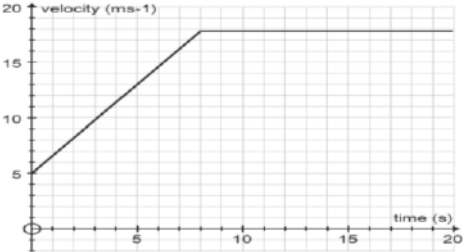
Question		Answer/Indicative content	Marks	Guidance	
26		DR $0.025t^3 - 0.8t^2 + 6.4t = 0$ $0.025t(t^2 - 32t + 256) = 0 \Rightarrow 0.025t(t - 16)^2 = 0$ $t = 0 \text{ or } 16$ Distance is $\int_0^{16} (0.025t^3 - 0.8t^2 + 6.4t) dt$ $= \left[0.025 \frac{t^4}{4} - 0.8 \frac{t^3}{3} + 6.4 \frac{t^2}{2} \right]_0^{16}$ $= \left(0.025 \times \frac{16^4}{4} - 0.8 \times \frac{16^3}{3} + 6.4 \times \frac{16^2}{2} \right) - (0)$ Distance = 137 m (3 sf)	M1(AO3.4) A1(AO2.1) M1(AO3.4) A1(AO1.1b) M1(AO2.1) B1(AO1.1b) [6]	Equating v to 0 for time at junction Factorising seen; method must be clear Limits not required for this mark Correct integration, and limits seen Use of limits seen; substitution of limits into integral must be seen Allow for any method seen	$\frac{2048}{5} - \frac{16384}{15} + \frac{4096}{5}$ $\frac{2048}{15} = 136.53...$
		Total	6		

Question			Answer/Indicative content	Marks	Guidance	
27		a	For AB: use of $s = ut + \frac{1}{2}at^2$ with $s = 64$, $t = 4$ gives $64 = 4u + \frac{1}{2}a \times 4^2$ For AC: $s = 96$ and $t = 8$ $96 = 8u + \frac{1}{2}a \times 8^2$ Solve simultaneously ($16 = u + 2a$, $12 = u + 4a$) $u = 20$ and $a = -2$ Alternative solution For AB: use of $s = ut + \frac{1}{2}at^2$ with $s = 64$, $t = 4$ gives $64 = 4u + \frac{1}{2}a \times 4^2$ For BC: speed at B is $u + 4a$ $32 = 4(u + 4a) + \frac{1}{2}a \times 4^2$ Solve simultaneously ($16 = u + 2a$, $8 = u + 6a$) $u = 20$ and $a = -2$	M1(AO3.3) A1(AO1.1b) B1(AO3.1b) M1(AO3.4) M1(AO1.1a) A1(AO1.1b) M1 A1 B1 M1 M1 A1 [6]	Correct (unsimplified) equation For both 96 and 8 seen Forming second equation in u and a May be implied if calculator used Both correct; allow deceleration = 2 Correct (unsimplified) equation Use of $v = u + at$ for AB Forming second equation in u and a oe BC Both correct; allow deceleration = 2	Allow use of a different sign convention (eg negative a) provided it is used consistently throughout and is explained Allow use of a different sign convention (eg negative a) provided it is used consistently throughout and is explained

Question			Answer/Indicative content	Marks	Guidance
		b	$u = 20, v = 0, a = -2$ gives $0 = 20^2 - 2 \times 2 \times s$ $s = 100$ so truck comes to rest 4 m beyond C	M1(AO3.4) A1(AO1.1b) [2]	Allow for any <i>suvat</i> equation(s) leading to a value for <i>s</i>
			Total	8	
28		a	$v^2 = u^2 + 2as$ $1.2^2 = 2 \times a \times 1.8$ $a = 0.4 \text{ ms}^{-2}$	M1 (AO 3.3) A1 (AO 1.1b) [2]	Using suitable <i>suvat</i> equation(s) leading to value for <i>a</i> <u>Examiner's Comments</u> Most candidates successfully used the <i>suvat</i> equations to reach the correct answer.
		b	$F - 19 = 2.8a$ $F = 20.12$ (20.1 N to 3sf)	M1 (AO 3.3) A1 (AO 1.1b) [2]	Using Newton's second law. All terms present. Allow 20 N FT their <i>a</i> <u>Examiner's Comments</u> Newton's second law was well understood and most candidates were successful here.
			Total	4	

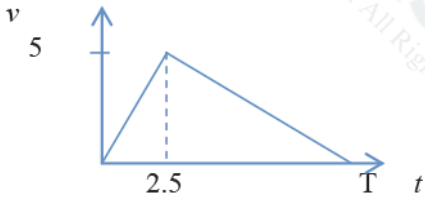
Question			Answer/Indicative content	Marks	Guidance
29		a	$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (12 - 2 \times 2t)\mathbf{i} + (2t - 6)\mathbf{j}$	M1 (AO 1.1a) A1 (AO 2.5) [2]	Attempt to differentiate at least one coefficient Must use vector notation Examiner's Comments Most candidates used vector notation accurately and successfully differentiated to obtain a correct expression for velocity.
		b	When $t = 3$ both components of velocity are zero, so the particle is stationary at $t = 3$.	M1 (AO 3.1a) E1 (AO 2.2a) [2]	Equating at least one component of their vector velocity to zero Must be argued from two zero components Examiner's Comments This was typically well answered with most candidates realising the requirement for both components to be zero at the same time.  Misconception It is not sufficient to equate the components and solve to find $t = 3$. From this starting point, candidates would need to check that the components were zero to achieve the method mark.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
30		a	$u = 5, v = 11.4, t = 4$ $a = \frac{v-u}{t} = \frac{11.4-5}{4} = 1.6$ $v = 5 + 1.6t$	M1 (AO 3.1b) A1 (AO 1.1b) A1 (AO 3.3) [3]	Using suvat equation(s) leading to value for a Any form FT their a <u>Examiner's Comments</u> The key to this question was to calculate the acceleration of the car. The required expression is then found by substituting the values for u and a into the equation $v = u + a t$. Many fully correct answers were seen.
		b	The car would not be able to accelerate indefinitely – the velocity would become too large	E1 (AO 3.5b) [1]	<u>Examiner's Comments</u> Candidates were required to recognise the limitations of this model. Most successful answers stated that the velocity would eventually get much too big or that the car would have to slow down or stop at some point.

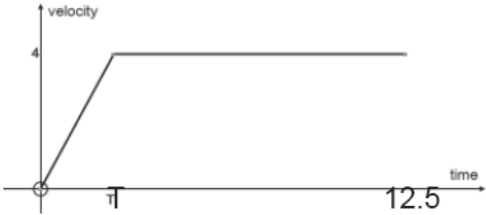
Question	Answer/Indicative content	Marks	Guidance
c	When $v = 17.8$ $t = \frac{17.8 - 5}{1.6} = 8$ 	B1 (AO 1.1a) G1 (AO1.1a) G1 (AO 3.5c) [3]	Calculation or point on graph labelled at $t = 8$ Two line segments with one horizontal Axes labelled. (0, 5) and constant speed 17.8 clear on vertical scale Mark intent for 17.8 – allow for a linear scale beyond 17.8 Examiner's Comments Most candidates correctly had a graph consisting of two line segments but a common error was to begin the graph at the origin when the initial velocity was 5 ms^{-1}. Some did not fully label their graph so lost a mark.  AfL Make sure the axes are labelled and that all key points are clearly indicated on the graph.
d	Dividing area into sections Area under trapezium $= \frac{1}{2}(5 + 17.8) \times 8 = 91.2$ Area rectangle $12 \times 17.8 = 213.6$ Total displacement = 304.8 m	M1 (AO 3.1b) A1 (AO 1.1a) A1 (AO 1.1b) [3]	May be found as sum of areas. May be implied by correct total FT their distance found for first 8s 213.6 must be added to another distance Examiner's Comments Many candidates were successful in finding the area under their graph with only a few arithmetical errors.

Question			Answer/Indicative content	Marks	Guidance	
		e	When $t = 4$ $v = 5 + 0.3 \times 4^2 - 0.05 \times 4^3 = 11.4 \text{ ms}^{-1}$ Which matches the given value	B1 (AO 3.4) [1]	Allow without comment	
		f	$\frac{dv}{dt} = 0.6 \times 2t - 0.05 \times 3t^2 \quad [=1.2t - 0.15t^2]$ When $t = 8$ $v = 1.2 \times 8 - 0.15 \times 64 = 0$ Acceleration is zero at $t = 8$ which means that the car reaches its maximum speed without the sudden change in acceleration in model A.	M1 (AO 1.1a) A1 (AO 3.2a) E1 (AO 3.2a) [3]	Need not be simplified Must mention acceleration Must compare with model A	Final mark can be awarded independently for a statement about change in acceleration as long as supported by some numerical evidence <u>Examiner's Comments</u> Some candidates were very vague in their answer here and were not credited many marks. There is a very clear instruction that it is the value of acceleration that is needed, so 2 of the 3 marks were given for finding this. Many candidates were able to comment that model B gives a gradual change in acceleration as it approaches the maximum speed avoiding the very sudden change seen in model A.

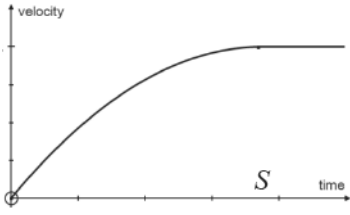
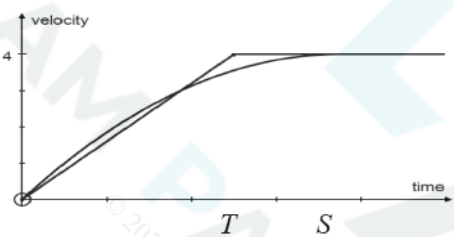
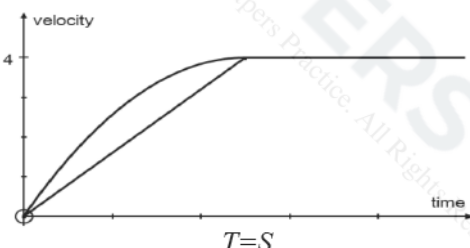
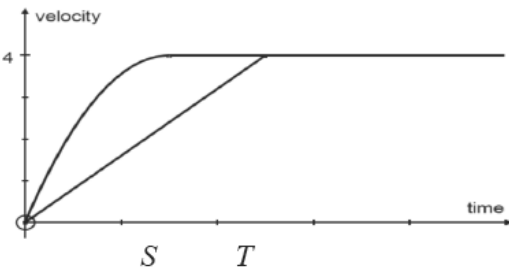
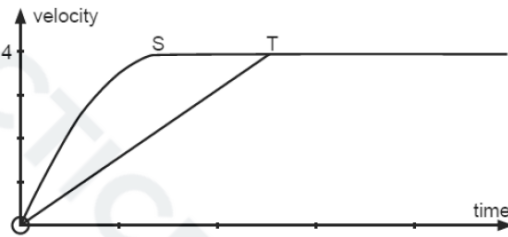
Question			Answer/Indicative content	Marks	Guidance	
		g	EITHER $\int_0^8 (5 + 0.6t^2 - 0.05t^3) dt = [5t + 0.2t^3 - 0.0125t^4]_0^8$ = 91.2 m which is same as model A for the first 8 s Distance is the same for the remainder of the time So this is the same as model A at $t = 20$ OR $\int_0^8 (5 + 0.6t^2 - 0.05t^3) dt = [5t + 0.2t^3 - 0.0125t^4]_0^8$ = 91.2 m Distance at 17.8 ms^{-1} 213.6 Total distance 304.8m [which is the same as model A]	M1 (AO 2.1) A1 (AO 1.1b) E1 (AO 2.1) [3] M1 A1 A1	BC Must consider to $t = 20$ BC Must consider to $t = 20$	Allow for correct definite integral stated and calculator used. Also allow M1A1 for $5 \times 8 - 0.2 \times 8^3 - 0.0125 \times 8^4$ seen Allow for correct definite integral stated and calculator used. Examiner's Comments Many candidates realised that the distance was the definite integral that gave the distance travelled in the first 8s. It would have been sufficient to clearly write the integral with its limits and use a calculator to evaluate it. Most candidates decided to give a full solution with the substitution of limits made clear. Only a few candidates omitted the part of the journey beyond 8s.
			Total	17		

Question	Answer/Indicative content	Marks	Guidance
31	EITHER acceleration phase $v^2 = 0 + 2.5 \times 2 = 5 \text{ m s}^{-1}$ slowing phase $v^2 = u^2 + 2as$ $0 = 5^2 + 2a \times 10$ $a = -1.25 \text{ m s}^{-2}$ $[-R] = 1.5 \times (-1.25) = -1.875$ Magnitude of $R = 1.875 \text{ N}$ (1.88 to 3sf) OR acceleration phase $v = 0 + 2.5 \times 2 = 5 \text{ m s}^{-1}$  using the distance to find the time it takes to stop <i>using areas</i> (second triangle): $\frac{1}{2}(T - 2.5) \times 5 = 10$ $T = 6.5 \text{ so time to stop is } 4 \text{ s.}$ So $0 = 5 + 4a$	M1 (AO3.1b) A1 (AO1.1b) M1 (AO3.1b) A1 (AO1.1b) M1 (AO1.1a) A1 (AO1.1b) [6] M1 A1 M1 A1	Use of <i>suvat</i> equation(s) to find velocity. Do not allow if $s = 10$ used Use of <i>suvat</i> equation(s) with $s = 10$ to find acceleration FT their velocity. Must be correct sign. Use of Newton's second law. FT their a $a \neq 2$ Must be positive Use of <i>suvat</i> equation(s) to find velocity. Do not allow if $s = 10$ used Use of area and <i>suvat</i> equation(s) to find acceleration Must be correct sign Must recognise two phases of motion for first 4 marks Consistent sign convention needed for full credit. Must recognise two phases of motion for first 4 marks

Question			Answer/Indicative content	Marks	Guidance	
			Giving $a = -1.25 \text{ m s}^{-2}$ $[-R] = 1.5 \times (-1.25) = -1.875$ Magnitude of $R = 1.875 \text{ N}$ (1.88 to 3sf)	M1 A1 [6]	Use of Newton's second law. FT their a $a \neq 2$ Must be positive <u>Examiner's Comments</u> Many good clear solutions were seen, however some candidates did not realise that this question covered two phases of motion and used all the numbers in the question in a single set of suvat equations. Some simply extracted the values of mass and acceleration from the first phase of motion and multiplied them together. Some candidates who obtained a negative value for resistance did not notice that it was the magnitude of the resistance that was required, so a positive answer was needed.  Look out for two phases of motion and set up different equations for the two phases. Use the value of the velocity at the end of the first phase to link the two phases.	Consistent sign convention needed for full credit.
			Total	6		

Question	Answer/Indicative content	Marks	Guidance
32	a 	B1 (AO1.1a) B1 (AO1.1a) [2]	Two line segments with one horizontal ($T, 4$) and ($12.5, 4$) labelled or indicated on scales. Allow their 2.5 marked instead of T. On axes labelled v and t or Examiner's Comments Most candidates gave a graph with two straight line segments but marks were often lost for graphs that were not fully labelled.  Make sure you label the axes and show the values of v and t at the significant points.
	b $\frac{1}{2} \times 4 \times (12.5 + (12.5 - T)) = 45$ $T = 2.5$	M1 (AO3.1a) A1 (AO1.1b) [2]	Attempt to find area of trapezium or both the triangle $\left(\frac{1}{2} T \times 4\right)$ and the rectangle $(12.5 - T) \times 4$. cao Examiner's Comments Candidates who used the area under the graph were usually successful. Candidates using the <i>suvat</i> equations often incorrectly combined values from the two separate phases of motion into a single equation. <i>Suvat</i> equations can be used for two phases of motion.

Question			Answer/Indicative content	Marks	Guidance
		c	EITHER $a = \frac{4}{2.5} = 1.6 \text{ m s}^{-2}$ $s = \frac{1}{2} \times 1.6t^2 = 0.8t^2$ OR $a = \frac{4}{2.5} = 1.6 \text{ m s}^{-2}$ $v = \int a \, dt = 1.6t + c$ When $t = 0$, $v = 0$ so $c = 0$ $s = \int v \, dt = 0.8t^2 + c$ When $t = 0$, $s = 0$ so $c = 0$ Giving $s = 0.8t^2$	M1 (AO1.1a) A1 (AO3.3) [2] M1 A1 [2]	Soi FT their T Soi FT their T Must be complete solution – do not award without consideration of + c at least once Examiner's Comments This was not well answered, as many candidates did not realise that the value of the acceleration was the key to this question. Many incorrectly used $s = \frac{1}{2}(u + v)t$ with $u = 0$ and $v = 4$, and the resulting linear expression did not qualify for follow-through marks in part (d).
		d	$0.8t^2 = 4$ $t = \sqrt{5} = 2.24 \text{ s}$	B1FT (AO3.4) [1]	FT their quadratic model in (c) Examiner's Comments This was usually credited to candidates who had had a quadratic expression for s in part (iii) as follow-through was allowed.

Question	Answer/Indicative content	Marks	Guidance
e		B1 (AO1.1a) [1]	Must have curved section of the graph decreasing gradient. S must be labelled. Examiner's Comments Most candidates who attempted this question got it right. The mark was only credited where the curved part of the graph had a decreasing gradient.
f	Total distance (area under the graph) can only be equal if $S > T$ If $S > T$  If $S = T$  If $S < T$ 	E1 (AO3.5c) [1]	Needs to give reason relating to the refinement of the model. Graphs not required Examiner's Comments Very few candidates were credited this mark. Many argued that the decreasing gradient implied it would take longer to reach maximum speed – the incorrect underlying assumption here was that the gradient at the origin would be the same, so these arguments were not credited the mark. This model gives the same total distance in 12.5 s and only answers which compared distances or areas were eligible for the mark. The easiest way to decide was to sketch the graphs with $S > T$ and to realise that this meant the total distance would be larger and so to argue that $S > T$. 
Total		9	

Question			Answer/Indicative content	Marks	Guidance
33		a		B1 (AO 3.3) B1 (AO 1.1) B1 (AO 1.1) [3]	Graph with three straight lines with positive, zero and negative gradients Vertical axis labelled with 2.5 and 7.5 marked Horizontal axis labelled with 12, 32 and 40 marked (oe, e.g. separate time intervals indicated)
		b	Total distance = area under graph $= \frac{1}{2} \times (2.5 + 7.5) \times 12 + (20 \times 7.5) + \frac{1}{2} \times 8 \times 7.5$ = 240 m Average speed is $\frac{240}{40}$ = 6 m s⁻¹	M1 (AO 3.1b) A1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) [4]	Any complete method Dividing their distance by their total time FT
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
34		a	$v = 0.1$ when $t = 1$ gives $a + b = 0.1$ $\frac{dv}{dt} = 4at^3 + 3bt^2$ Maximum v when $t = 1$ gives $4a + 3b = 0$ Solving simultaneous equation for a and b $a = -0.3$ and $b = 0.4$	M1 (AO 3.3) M1 (AO 3.1b) M1 (AO 3.3) M1 (AO 1.1a) A1 (AO 1.1) [5]	Using given information to find an equation linking a and b Equating the derivative to zero to find an equation linking a and b Method may be implied, e.g. if BC cao
		b	Changes direction when $v = 0$, so $-0.3t^4 + 0.4t^3 = 0 \Rightarrow 0.3t^4 \Rightarrow 0.4t^3 \Rightarrow t = \frac{4}{3}$ (as $t > 0$) Particle changes direction when $t = \frac{4}{3}$	M1 (AO 3.4) A1 (AO 1.1) [2]	For equating v to 0 and solving for t (may be BC); ignore any inclusion of $t = 0$ at this point cao
		c	Model is not suitable for large values of t as the object's velocity would increase without limit	B1 (AO 3.5b) [1]	oe, e.g. 'velocity gets very large'
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
35		a	Resolve perpendicular to plane: $R = 2.7g \cos 25^\circ$ so $F = \mu R$ gives $F = 0.4 \times 2.7g \cos 25^\circ$ $(= 9.59\dots)$ Newton II $\parallel$ to plane: $24 - F - 2.7g \sin 25^\circ = 2.7a$ $a = 1.1945\dots$ so acceleration is 1.19 m s^{-2} (3sf)	B1 (AO 3.3) M1 (AO 3.4) A1 (AO 1.1) M1 (AO 3.3) A1 (AO 1.1) [5]	Allow sin/cos interchange if consistent error elsewhere For $0.4 \times$ their R, but not if $R = 2.7g$ For correct expression for F; evaluation not needed here All forces needed and the 'weight' term must be a component; allow sign errors awrt 1.19
		b	After 5 s, $v = 0 + 1.1945 \times 5$ $v = 5.9725$ Newton II $\parallel$ to plane: $-F - 2.7g \sin 25^\circ = 2.7a$ $a = -7.6943$ $v = 0$ at distance s where $0 = 5.9725^2 - 2 \times 7.6943 \times s$ $s = 2.3179\dots$ so distance travelled is 2.32 m (3sf)	M1 (AO 3.1b) A1 (AO 1.1) M1 (AO 3.3) A1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) [6]	Use of <i>suvat</i> leading to a value for v Both forces, with weight resolved Use of <i>suvat</i> leading to a value for s cao
			Total	11	

Question			Answer/Indicative content	Marks	Guidance	
36		a	$4\mathbf{i} + 6\mathbf{j} = -2\mathbf{j} + 8\mathbf{a}$ $\mathbf{a} = 0.5\mathbf{i} + \mathbf{j} \text{ m s}^{-2}$	M1 (AO 1.1a) A1 (AO 2.5) [2]	For use of vector equation $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, or alternatively for finding both acceleration components separately Must be vector form; do not allow final answer as separate components	Do not allow for the magnitude of the acceleration unless final answer is labelled as such
		b	$\mathbf{v} = -2\mathbf{j} + (0.5\mathbf{i} + \mathbf{j})t$ is directed east when $-2 + t = 0$ $t = 2$, so boat sails east at time 2 s	M1 (AO 3.1b) A1 (AO 3.2a) [2]	For equating j component of velocity to zero, giving equation for t cao	
		c	Displacement in 4 s is $(-2\mathbf{j}) \times 4 + \frac{1}{2}(0.5\mathbf{i} + \mathbf{j}) \times 4^2$ $= 4\mathbf{i} (+ 0\mathbf{j})$ Position at time $t = 4$ gives $\mathbf{r}_A + 4\mathbf{i} = 5\mathbf{i} - 2\mathbf{j}$ $\mathbf{r}_A = \mathbf{i} - 2\mathbf{j}$	M1 (AO 3.1b) A1 (AO 1.1) M1 (AO 3.1b) A1 (AO 1.1) [4]	For any use of $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ with $t = 4$ and their $\mathbf{a}$ even if no clarity about displacement/ position For correct use of position vectors Must be in vector form	Equation may include initial position term $\mathbf{r}_A$ May be earned earlier, if original equation is $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$
			Total	8		

Question			Answer/Indicative content	Marks	Guidance	
37			Use of $s = ut + \frac{1}{2}at^2$ to compare two accelerations For OA: $24 = 3 \times 4 + \frac{1}{2} a \times 4^2$ $a = 1.5$ For OB: $104 = 3 \times 10 + \frac{1}{2} a \times 10^2$ $a = 1.48$ Similar values, so constant acceleration is a good model Alternative solution Predicting a value and comparing with given figure For OA: $24 = 3 \times 4 + \frac{1}{2} a \times 4^2$ $a = 1.5$ For OB: $s = 3 \times 10 + \frac{1}{2} \times 1.5 \times 10^2$ OB = 105 m Actual distance is 104 m, which is very close, so constant acceleration is a good model	M1 (AO 3.3) A1 (AO 1.1b) M1 (AO 3.3) A1 (AO 1.1b) A1 (AO 3.5a) M1 A1 M1 A1 A1 [5]	Use formula with $u = 3, s = 24, t = 4$ Use formula with $u = 3, s = 104, t = 10$ or (for AB) with $u = 9, s = 80, t = 6$ (or $a = 1.44$ using data for AB) Clear comparison and conclusion. Allow alternative conclusion, i.e. that the model is not [exactly] consistent with the data Find a for OA using $u = 3, s = 24, t = 4$ Use of $a = 1.5$ for OB, oe Clear comparison and conclusion. Allow alternative conclusion, i.e. that the model is not [exactly] consistent with the data	If AB considered do not allow $u = 3$; there must be an attempt to find the speed at A, e.g. via <i>suvat</i> for OA Allow credit for any complete method Do not allow $u = 3$ as initial speed for AB
			Total	5		

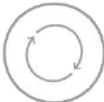
Question			Answer/Indicative content	Marks	Guidance	
38		a	Two valid comments, e.g.: The car's direction of motion changes from negative to positive [at time $t = 1$] The initial speed of the car is 1 m s^{-1} [in the negative direction] After 1 s the car is momentarily stationary The car accelerates [in the positive direction] reaching a speed of 19.8 m s^{-1} [after 4 seconds]	B1 (AO 2.2a) B1 (AO 2.2a) [2]	For a comment involving the change of direction For any essentially different sensible comment about the motion	
		b	DR Consideration of two separate phases of the motion $s = \int (0.1t^3 + 0.9t^2 - 1) dt = 0.025t^4 + 0.3t^3 - t (+c)$ For 1st second: $s = (0.025 \times 1^4 + 0.3 \times 1^3 - 1) - 0 = -0.675$ For the next three seconds: $s = (0.025 \times 4^4 + 0.3 \times 4^3 - 4) - (-0.675)$ $= 22.275$ Total distance = $22.275 + 0.675 = 22.95\text{m}$	M1 (AO 3.1b) M1 (AO 1.1a) A1 (AO 1.1b) M1 (AO 1.1a) B1 (AO 1.1b) [5]	May be implied Attempt to integrate the terms is needed Correct indefinite integration (may be seen as working for a definite integral) For substitution of limits, oe Allow for correct answer seen, www	+ c not required here Allow this mark for limits 0 and 4
		c	$a = \frac{dv}{dt} = 0.3t^2 + 1.8t$	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Attempt to differentiate cao	
			Total	9		

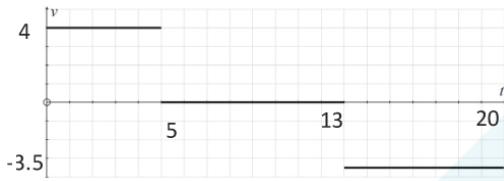
Question			Answer/Indicative content	Marks	Guidance	
39		a	Use of Newton's 2nd Law for the whole train Weight component parallel to the plane is $0.5g \sin 20^\circ$ $2.5 - 0.5g \sin 20^\circ = 0.5a$ $a = 1.65 \text{ ms}^{-2}$ to 3sf Alternative solution Equations for engine and truck separately N2L for engine: $2.5 - 0.3g \sin 20^\circ - T = 0.3a$ N2L for truck: $T - 0.2g \sin 20^\circ = 0.2a$ $a = 1.65 \text{ ms}^{-2}$ to 3sf	B1 (AO 3.3) M1 (AO 1.1a) A1 (AO 2.1) B1 M1 A1 [3]	May be implied by use in an equation N2L with at least one correct force AG; must be clearly shown Correct weight component in at least one equation Both equations attempted, plus attempt to eliminate T AG; must be clearly shown	$m = 0.5$ used $m = 0.3$ and/or 0.2 used
		b	Either $T - 0.2g \sin^\circ = 0.2 \times 1.65$ Or $2.5 - 0.3g \sin 20^\circ - T = 0.3 \times 1.65$ $T = 1 \text{ N}$	M1 (AO 3.3) A1 (AO 1.1b) [2]	Use of N2L for either engine or truck Allow awrt 1.00	Equation must have all the forces and no extras
		c	$3^2 = 1^2 + 2 \times 1.65 \times s$ $s = 2.42 \text{ m}$	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Use of suvat leading to a value for s Or 2.43 from use of $a = 1.6482...$	
			Total	7		

Question			Answer/Indicative content	Marks	Guidance
40	a		DR Using given values in $v = 1.2t^2 - kt^3$: $25.6 = 1.2 \times 8^2 - k \times 8^3$ $k = \frac{25.6 - 76.8}{-8^3} = 0.1$	M1 (AO3.3) A1 (AO2.1) [2]	method must be clear AG Allow M1 for substitution of $t = 8$, $k = 0.1$ to obtain $v = 25.6$ A1 for comment made that $k = 0.1$ is correct value. <u>Examiner's Comments</u> This was generally well answered by candidates who substituted the values of velocity and time and rearranged establish the correct value for k. Some candidates used the given value for k as well to obtain the value for v, but this was only awarded one mark unless a conclusion about k was also seen.
	b		DR Uses $v = u + at$ at $u = 0$, $v = 25.6$, $t = 8$ $\Rightarrow 25.6 = 0 + 8a$ $a = 3.2 \text{ ms}^{-2}$	M1 (AO3.3) A1 (AO1.1) [2]	Cao <u>Examiner's Comments</u> Many candidates did not pick up that the model had changed and so differentiated the first model instead of using the constant acceleration formulae. Those using $v = u + at$ were usually successful in obtaining both marks.  AfL Candidates should be aware that where there is a line of text that is not indented between part (a) and part (b), the situation will have changed.

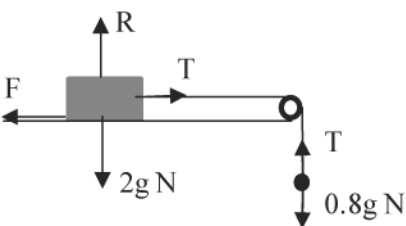
Question		Answer/Indicative content	Marks	Guidance	
	c	DR First model $s = \int_0^8 (1.2t^2 - 0.1t^3) dt = [0.4t^3 - 0.025t^4]_0^8$ $(0.4 \times 8^3 - 0.025 \times 8^4) - (0.4 \times 0 - 0.025 \times 0)$ $= 102.4 \text{ m}$ Second model: use <i>suvat</i> equation(s) with 3 from $u = 0$, $v = 25.6$, $t = 8$, $a =$ their value, giving e.g. $s = 0 \times 8 + \frac{1}{2} \times 3.2 \times 8^2 \text{ or}$ $s = \frac{1}{2} (0 + 25.6) 8$ $= 102.4 \text{ m, [which is the same value as for 1st model]}$	M1 (AO3.4) M1 (AO1.1) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) [5]	Attempt to integrate – expression for indefinite integral must be seen Both limits seen or establishes that $c = 0$ if indefinite integral used. Allow following first M mark Method must be clear Allow without comment Examiner's Comments This question required candidates to find displacement for model 1 using integration and for model 2 using constant acceleration. A complete solution should also include a comment that the two answers were the same, but full credit was awarded for two correct answers obtained from the two models. To obtain full marks, a definite integral using the limits of 0 and 8 was required, or the candidate had to establish the value of the constant of integration to be zero. Many candidates only used one of the two models. Some used the constant acceleration formulae in two different ways.	Allow SC1 for $\int_0^8 (1.2t^2 - 0.1t^3) dt = 102.4$
		Total	9		

Question			Answer/Indicative content	Marks	Guidance	
41			N2L gives $4000 - 800 = 1200a$ $a = \frac{8}{3} \text{ m s}^{-2}$ Using $v = u + at$ gives $v = \frac{8}{3} \times 9 = 24 \text{ m s}^{-1}$	M1 (AO 3.1b) A1 (AO 1.1b) M1 (AO 3.3) A1 (AO 1.1b) [4]	Both forces present; allow sign errors Soi Using <i>suvat</i> equation(s) leading to a value for v FT their a Examiner's Comments A few candidates were not clear about the distinction between mass and weight and used $F = mga$ for Newton's second law. The majority of candidates knew how to calculate the acceleration and use it to find the velocity. Some rounded or truncated their acceleration, but rarely so much that their final answer did not agree with the given answer to at least 2 significant figures.	Allow for $3200 = 1200a$ seen
			Total	4		

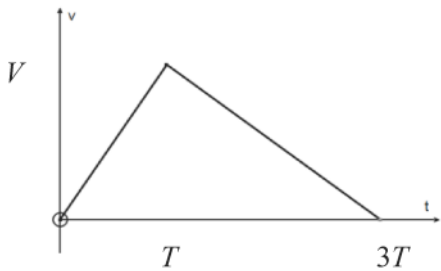
Question		Answer/Indicative content	Marks	Guidance																
42		$v = 3.5t - 0.5t^2 \Rightarrow \frac{dv}{dt} = 3.5 - t$ When $t = 8$, $a = 3.5 - 8 = -4.5$ And $v = 3.5 \times 8 - 0.5 \times 8^2 = -4$ Either “Velocity is negative and decreasing, so the speed is increasing”. Or “As both the velocity and acceleration are negative, the speed is increasing” or similar Special case When $t = 8$, $v = 3.5 \times 8 - 0.5 \times 8^2 = -4$ Evaluating v either side of $t = 8$ Statement referring to correct values <table><tr><td>t</td><td>7</td><td>7.5</td><td>7.9</td><td>8</td><td>8.1</td><td>8.5</td><td>9</td></tr><tr><td>v</td><td>0</td><td>-1.875</td><td>-3.55</td><td>-4</td><td>-4.45</td><td>-5.88</td><td>-9</td></tr></table>	t	7	7.5	7.9	8	8.1	8.5	9	v	0	-1.875	-3.55	-4	-4.45	-5.88	-9	M1 (AO 3.1b) A1 (AO 2.1) B1 (AO 3.1b) A1 (AO 3.2a) [4] B1 B1 B1	Attempt to differentiate to find a Evaluating a when $t = 8$ Evaluating v when $t = 8$ Argued from negative a and v Evaluating v when $t = 8$ Two correct values Argued from correct working Examiner’s Comments Many candidates correctly evaluated acceleration to show that the velocity is decreasing and did not consider the value of velocity, or the concept of speed, at all. Others who had the negative values for v and a did not complete their argument. Candidates who used a good sketch graph of velocity were almost all successful here. Some candidates wasted time here by using the product rule for differentiation rather than expanding the brackets.  Misconception Candidates did not appear to understand the distinction between velocity and speed.  AfL The command word ‘Determine’ is not an
t	7	7.5	7.9	8	8.1	8.5	9													
v	0	-1.875	-3.55	-4	-4.45	-5.88	-9													

Question			Answer/Indicative content	Marks	Guidance	
					instruction prohibiting the use of the numerical differentiation function on the calculator but it does indicate that an explicit line of reasoning must be seen to show that a is found by $a = \frac{dv}{dt}$.	
			Total	4		
43	a			B1(AO1.1) B1(1.1) [2]	Graph from 3 horizontal line segments. Correct velocities labelled Times seen – either $t = 5, 13, 20$ or lengths of line segments 5, 8, 7 seen.	Any lines joining the horizontal lines should be vertical
	b		Distance = $(4 \times 5) + (7 \times 3.5)$ m =44.5	M1(AO1.1a) A1(AO1.1) [2]	finding the area of at least one region from their graph or cao	May work directly from the information in the question without reference to their graph
	c		Displacement = $20 - 24.5 = -4.5$ m	B1(AO1.1) [1]	Allow for -4.5 m or for 4.5 m south	Do not allow -4.5 m south
			Total	5		

Question			Answer/Indicative content	Marks	Guidance	
			So v is an increasing function When $t = 0$, $v = 3 > 0$ $v[> 3] > 0$ for all values of t So the velocity is never zero and the particle never stationary.		Attempt to construct an argument based on the positivity of v. Uses the positivity of t aiming to establish the positivity of a Clear argument that $a > 0$ Uses the link between $a > 0$ and v Uses $v_0 = 3$ explicitly in their argument Convincing complete argument.	
			Total	6		

Question			Answer/Indicative content	Marks	Guidance	
45	a			B1(1.1a) B1(1.1a) B1(1.1a) [3]	both weights correct. common tension in the right directions Friction and normal reaction and no extra forces	Allow weight of box and weight of sphere, but not if both marked weight. Allow T_1 and T_2 provided they are clearly shown equal to each other elsewhere. For F, allow $0.35R$, $0.35 \times 2g$ $0.7g$ or 6.86 N
	b		$T - F = 2a$ $0.8g - T = 0.8a$	B1(AO1.1a) B1(1.1a) [2]	Allow any expression for F Allow distinct tensions if consistent with diagram	For F, allow $0.35R$, $0.35 \times 2g$ $0.7g$ or 6.86 N
	c		Vertically for the block $R = 2g$ Friction $F = 0.35R = 0.7g$ Add equations $0.8g - F = 2.8a$ $0.8g - 0.7g = 2.8a$ $a = 0.35\text{ m s}^{-2}$	M1(AO3.1b) A1(AO2.1) M1(AO2.1) A1(2.1) [4]	Attempt to use μ to evaluate friction Correct value for F Eliminate T from their equations AG must follow from correct work	Some of this work may already have been seen in previous part
	d		Use $s = 0.5$, $u = 0$, $a = 0.35$ $0.5 = \frac{1}{2} \times 0.35 \times t^2$ $t = 1.69\text{ s}$	M1(AO1.1a) A1(AO1.1) [2]	Using $suvat$ equation(s) leading to a value for t Do not allow ± 1.69	
			Total	11		
46			eg. Neglect air resistance Constant gravity Projectile is a particle	B1(AO1.2) [1]	One sensible statement.	Do not accept level ground
			Total	1		

Question			Answer/Indicative content	Marks	Guidance	
47	a		$v = t^3 - 5t^2 = 0$ $t^3(t - 5) = 0$ So at rest when $t = 0$ or 5 s	M1(AO1.1a) A1(AO1.1) [2]	DR Attempt to solve $v = 0$ Both roots and no others	Allow M1 for at least one root found by substitution
	b		There is a change of direction when $t = 5$ $\int_0^5 (t^3 - 5t^2) dt = \left[\frac{t^4}{4} - \frac{5t^3}{3} \right]_0^5 = -\frac{625}{12}$ $\int_5^6 (t^3 - 5t^2) dt = \frac{193}{12}$ Total distance $\frac{625 + 193}{12} = \frac{409}{6} = 68.2$ m	M1(AO3.4) M1(AO3.4) A1(AO1.1) [3]	DR Considering signed areas either side of $t = 5$ s. Algebraic integration seen attempted Correct to at least 2 sf	SPECIAL CASE $\int_0^6 (t^3 - 5t^2) dt$ $= \left[\frac{t^4}{4} - \frac{5t^3}{3} \right]_0^6 = -36$ SCM1 for algebraic integration seen attempted SCA1 for -36 m seen or distance 36 given
			Total	4		

Question			Answer/Indicative content	Marks	Guidance	
48	a			B1(AO1.1a) B1(AO1.1a) [2]	Two line segments starting and ending on the x-axis T and 3T seen on t-axis	Instead of 3T, allow for 2T shown on the horizontal axis for the second phase
	b		Acceleration phase $\frac{V}{T} = 2.5$ Area under graph $\frac{1}{2} \times V \times 3T = 240$ Solving simultaneously $\frac{3}{2}(2.5T)T = 240 \Rightarrow T^2 = 64$ So $T = 8$ and $V = 20$ OR Acceleration phase using $v = u + at$ $V = 2.5 \times T$ First phase has distance $240 \div 3 = 80$ m Using $s = 80, u = 0, a = 2.5$ $80 = \frac{1}{2} \times 2.5T^2$ giving $T = 8$ and $V = 20$	B1(AO3.4) B1(AO3.4) M1(AO1.1a) A1(AO1.1) [4] B1 M1 B1 A1	soi May be sum of two areas soi Attempt to eliminate one variable correct pair of answers May be final step using $T = 8$ award if seen Using <i>suvat</i> equation(s) leading to a value for t with $s = 80$ correct pair of answers	$\frac{1}{2}VT + VT = 240$
			Total	4		

Question			Answer/Indicative content	Marks	Guidance	
49	a		Acceleration due to gravity is not a constant but depends on location in the universe.	B1(AO3.5b) [1]	Allow any sensible comment that g might be different on the moon	
	b		Using $s = 2$, $u = 0$, $t = 1.6$ and $s = ut + \frac{1}{2}at^2$ Downwards the positive direction $2 = \frac{1}{2} \times 1.6^2 a$ giving [$a = 1.5625$] which 1.6 m s ⁻² to 2 sf	M1(AO3.3) A1(AO1.1) [2]	Allow any sign convention Allow -1.6 m s ⁻² if upwards is clearly indicated as positive. Must be 2 significant figures.	
	c		Using $u = 15$, $v = 0$, $a = -1.6$ and $v^2 = u^2 + 2as$ Upwards the positive direction $0 = 15^2 - 2 \times 1.6s$ $s = 70.3$ m	M1(AO3.4) A1(AO1.1) [2]	Use of <i>suvat</i> equation(s) leading to a value for s . Allow sign errors Must follow from correct working and if negative, explained.	Allow answers in the range 70 to 72.2
			Total	12		